

Rhyme: A Fiat-Shamir Lattice-based Signature with 3C Sampling

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Abstract. In this paper, we propose a new postquantum lattice-based digital signature named Rhyme. The scheme is based on the Fiat-Shamir structure and does not rely on flooding, rejection sampling, or Gaussian convolution as previous methods. Instead, its security is based on a variant of LWE combined with a new sampling method (3C sampling). We prove its security in ROM/QROM and provide concrete parameters as well as reference implementation to show that our scheme enjoys high efficiency and very compact signature size compared with former results.

Key words: Lattice, LWE, Fiat-Shamir, Digital Signature

1 Introduction

With the rapid developments in quantum algorithms and computations, research in lattice-based cryptography has attracted considerable attention because lattice-based cryptosystems are likely to be effective against quantum computing attacks in the future. Schemes with various features, such as digital signatures, identity-based and attribute-based encryption, zero-knowledge proof and fully homomorphic schemes, can be realized based on the mathematical and computational properties of lattices. As a result, lattice-based cryptosystems are now regarded as promising candidates for post-quantum cryptography standardization process, e.g. NIST and KpqC.

Most lattice-based signatures are designed on the basis of three general structures, namely the GGH structure, the Fiat-Shamir structure, and the GPV trapdoor structure. Among them, the Fiat-Shamir structure enjoys a good balance of efficiency and signature size. As a result, CRYSTALS-Dilithium [31] and

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HAETAE [12] are separately chosen as the selection for standardization by NIST and KpqC, which are all based on Fiat-Shamir structure.

Lattice-based Fiat-Shamir structure is first used to design practical lattice-based signature scheme in [29]. This work combines the Fiat-Shamir structure with rejection sampling technology to avoid the risk of secret leakage. Many variants have been proposed based on this work including [2, 7, 25, 30, 31].

In designing Lattice-based signature based on Fiat-Shamir structure, a core problem is how to deal with the leakage of the secret key from signatures. First introduced in [29] and then further improved in [30], rejection sampling is an important technology to address this problem. The idea of rejection sampling is simple but effective, where it demands the signer selectively outputs signatures to ensure that the secret key should not be leaked by signatures. To achieve this goal, the rejection sampling process will choose to output a signature or reject it according to a fixed condition. After performing rejection sampling, the outputted distribution of signatures is adjusted to a certain target distribution that can be publicly sampled which ensures that no information of the secret key will be revealed. In rejection sampling, it is important to choose proper source and target distributions because the range of the target distribution determines the size of signature and the distance between the target and the source distributions determines the probability of rejection, thus influences the efficiency of sign process. Many schemes manage to lower the signature size by changing distributions of rejection sampling, e.g. HAETAE [12] uses uniform distributions of hyperballs to achieve smaller signature compared with CRYSTALS-Dilithium which uses bounded uniform distribution. Gärtnner introduces iterative rejection sampling to decrease the rejection probability and provides more flexibilities for choosing parameters of small signature size [26].

Recently, another framework to prevent the leakage of the secret key in the Fiat-Shamir structure has been proposed in G+G scheme [16]. In this scheme, the signature is made up of two parts which separately follows some Gaussian distributions. According to the convolution theorem of Gaussian distribution, the distribution of the outputted signature is statistically close to another Gaussian distribution under certain condition. Thus there is no need for rejection sampling in this scheme which results in a more compact signature size. However, to satisfy the condition of convolution theorem, the lower bound of the variance for the outputted distribution, which is decided by smoothing parameter, cannot be very small. This becomes the biggest limitation of further compressing the signature size.

Inspired by G+G, we propose another lattice-based Fiat-Shamir signature called Rhyme, which does not rely on rejection sampling or convolution theorem. Instead, our idea is built up of a basic observation that for any symmetric distribution ψ with the support $\mathbb{Z}$ and the center 0, $\psi = -\psi$ which always holds

due to the definition. Then we extend this observation to the convolutions of a number of symmetric distributions which leads to a new sampling method called convolution of constant count sampling (3C sampling). In 3C sampling, it is interesting to see that the output distribution is naturally independent of secret keys. Thus we change the distribution of secret key in the keygen process to satisfy the requirements of 3C sampling and prove this change also ensures the safety against the key recovery attack at the same time. At last, the new keygen process combined with 3C sampling method under Fiat-Shamir structure forms the Rhyme signature scheme.

Rhyme differs from the former results in three major aspects: 1) we use a variant of LWE problem for the key generation process of the signature, and give a reduction to proof its security. 2) we use a new method to sample mask vectors in Fiat-Shamir Signature denoted as Convolution of Constant Counts sampling (3C sampling) whose output distribution is naturally independent of the secret key. Combining this sampling method with the proposed new key generation process, we show that the scheme can hide the information of the secrets without using rejection sampling or flooding or Gaussian convolution theorem. Thus this property leads to a very compact signature size compared with former results. 3) Our scheme does not need float-point sampling like HAETAE or G+G but only integer sampling, so it is easy to be realized for both software and hardware. And we also provide a reference implementation for validation.

2 Preliminaries

2.1 Notations

Matrices are denoted in bold font and upper case letters (e.g., $\mathbf{A}$), while vectors (every vector is column vector) are denoted in bold font and lowercase letters (e.g., $\mathbf{y}$ or $\mathbf{z}_1$). The i -th component of a vector is denoted with subscript i (e.g., y_i for the i -th component of $\mathbf{y}$). We denote concatenation between vectors by putting the rows below as $(\mathbf{u}, \mathbf{v})$ $[[\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} ?]]$ and the columns on the right as $(\mathbf{u}|\mathbf{v})$. We naturally extend the latter notation to concatenations between matrices and vectors (e.g., $(\mathbf{A}|\mathbf{b})$ or $(\mathbf{A}|\mathbf{B})$). We let $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ be a polynomial ring where n is a power of 2 integer and for any positive integer q the quotient ring $\mathcal{R}_q = \mathbb{Z}[x]/(q, x^n + 1) = \mathbb{Z}_q[x]/(x^n + 1)$. We abuse notations and identify $\mathcal{R}_2$ with the set of elements in $\mathcal{R}$ with binary coefficients. Given $\mathbf{y} = (\sum_{0 \leq i < n} y_i x^i, \dots, \sum_{0 \leq i < n} y_{nk-n+i} x^i)^T \in \mathcal{R}^k$, we define its l_2 -norm as the l_2 -norm of the corresponding "flattened" vector $\|\mathbf{y}\|_2 = \|(y_0, \dots, y_{nk-1})^T\|_2$.

For a positive integer α , we define $r \bmod^\pm \alpha$ as the unique integer r' in the range $[-\alpha/2, \alpha/2)$ satisfying the relation $r = r' \bmod \alpha$. We also define $r \bmod^+ \alpha$ as the unique integer r' in the range $[0, \alpha)$ that satisfies $r = r' \bmod \alpha$. We

naturally extend this to integer polynomials and vectors of integer polynomials, by applying it component-wise.

2.2 Lattice

An m -dimensional *lattice* is a discrete additive subgroup in $\mathbb{R}^m$ which can be represented as the set of integer linear combination of n linearly independent vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, i.e.

$$\mathcal{L}(\mathbf{B}) = \left\{ \sum_{i=1}^n x_i \mathbf{b}_i \mid x_i \in \mathbb{Z}, \forall i \in [1, n] \right\}, \quad (1)$$

where $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_n]$ is called a *basis* of $\mathcal{L}$ which is not unique, $n(n \leq m)$ is the *rank* of the lattice, a lattice is called full-rank if $m = n$. The determinant of $\mathcal{L}$ is defined as

$$\det(\mathcal{L}) = \sqrt{\det(\mathbf{B}^\top \mathbf{B})}. \quad (2)$$

The quantity $\det(\mathcal{L})$ is invariant regardless of the choice of $\mathbf{B}$. The *dual lattice* $\mathcal{L}^*$ is defined as

$$\mathcal{L}^* = \{\mathbf{w} \in \mathbb{R}^m \mid \forall \mathbf{v} \in \mathcal{L}, \langle \mathbf{w}, \mathbf{v} \rangle \in \mathbb{Z}\}. \quad (3)$$

q-ary lattice As a kind of important lattices in lattice-based cryptography, a q -ary lattice refers to the lattice such that $q\mathbb{Z}^n \subseteq \mathcal{L} \subseteq \mathbb{Z}^n$ where q is an integer.

Two types of q -ary lattices frequently used in lattice cryptography are defined as follows with respect to an $n \times m$ matrix $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$,

$$\mathcal{L}_q(\mathbf{B}) = \{\mathbf{y} \in \mathbb{Z}^m \mid \mathbf{y} = \mathbf{B}^\top \mathbf{x} \bmod q, \mathbf{x} \in \mathbb{Z}^n\}, \quad (4)$$

$$\mathcal{L}_q^\perp(\mathbf{B}) = \{\mathbf{y} \in \mathbb{Z}^m \mid \mathbf{B}\mathbf{y} = 0 \bmod q\}. \quad (5)$$

2.3 LWE Problem

Learning with error (LWE) problem was proposed by Regev [37] in 2005 and has been widely used in the construction of lattice-based cryptography. We first introduce some definitions in order to describe LWE problems.

Definition 2.1 (LWE distribution). Let $n \geq 1$, $q \geq 2$ and χ be an error distribution over $\mathbb{Z}_q$, given a secret vector $\mathbf{s} \in \mathbb{Z}_q^n$, the LWE distribution $L_{\mathbf{s}, \chi}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing $\mathbf{a} \xleftarrow{\$} U(\mathbb{Z}_q^n)$ and $e \xleftarrow{\$} \chi$, and outputting $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$.

The LWE problem has a search version and a decision version, which are defined as follows.

Definition 2.2 (Search-LWE). Given m samples $(\mathbf{a}_i, b_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ that are independently sampled from $L_{\mathbf{s}, \chi}$ with a fixed secret $\mathbf{s} \in \mathbb{Z}_q^n$, the goal of search-LWE is to find the secret vector $\mathbf{s}$.

In the following part of this paper, we denote $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ to be the matrix formed by m columns $\{\mathbf{a}_i\}_{i=1}^m$ and $\mathbf{b} = (b_1, b_2, \dots, b_m)^\top \in \mathbb{Z}_q^m$, where $b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q$.

Definition 2.3 (Decision-LWE). *Given m independent samples $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{n \times m} \times \mathbb{Z}_q^m$ that follow either the LWE distribution $L_{\mathbf{s}, \chi}$ with a fixed secret $\mathbf{s} \in \mathbb{Z}_q^n$ or the uniform distribution, the goal of decision-LWE is to decide which distribution the samples follow.*

To make LWE more practical in cryptography, variants of LWE problems (e.g., Ring-LWE and Module-LWE) have been investigated. More details of these variants can be found in [32].

Definition 2.4 (Binary-LWE [9]). *Let $n \geq 1$, $q \geq 2$ and χ be an error distribution uniformly sampled from $\{0, 1\}$, given a secret vector $\mathbf{s} \in \mathbb{Z}_q^n$, the binary-LWE distribution $L_{\mathbf{s}, \chi}$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is sampled by choosing $\mathbf{a} \xleftarrow{\$} U(\mathbb{Z}_q^n)$ and $s, e \xleftarrow{\$} \chi$, and outputting $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$.*

2.4 Reduction from $LWE_{m,n,\sigma',q}$ to $LWE_{m,n,\sigma_k,q}$

In our scheme, we choose secret and error uniformly from $\{-1, 1\}$, denoted as σ_1 , it is not the usual case for LWE designs, so we will discuss its security by providing a reduction from $LWE_{m,n,\sigma',q}$ to $LWE_{m,n,\sigma_k,q}$ where σ_k means a distribution uniformly chosen from $\{-k, k\}$ and σ' is a uniform distribution in $\{0, 1\}$, whose corresponding problem is known as binary-LWE.

Consider a binary-LWE instance $LWE_{m,n,\sigma',q}$, $\mathbf{b} \in \mathbb{Z}_q^m = \mathbf{A}^\top \mathbf{s} + \mathbf{e} \bmod q$, where $(\mathbf{s}, \mathbf{e}) \in (\mathbb{Z}^n, \mathbb{Z}^m)$ are sampled from σ' . For some non-zero integer $k \in \mathbb{Z}_q$, we have $2k\mathbf{b} = 2k\mathbf{A}^\top \mathbf{s} + 2k\mathbf{e} = \mathbf{A}^\top (2k\mathbf{s}) + 2k\mathbf{e}$, let $\mathbf{s}' = 2k\mathbf{s}$, $\mathbf{e}' = 2k\mathbf{e}$ and $\mathbf{b}' = 2k\mathbf{b}$, so $\mathbf{b}' = \mathbf{A}^\top \mathbf{s}' + \mathbf{e}'$ where $\mathbf{s}', \mathbf{e}'$ follow a uniform distribution in $\{0, 2k\}$.

Let $\tilde{b}_i = b'_i - k \cdot \sum_{j=1}^n a_{j,i} - k = \sum_{j=1}^n a_{j,i} \cdot (s'_j - k) + e'_i - k$, then we have $\tilde{s}_j = s'_j - k$, $\tilde{e}_i = e'_i - k$. So the instance can be transformed to $\tilde{\mathbf{b}} \in \mathbb{Z}_q^m = \mathbf{A}^\top \tilde{\mathbf{s}} + \tilde{\mathbf{e}} \bmod q$ where $\tilde{\mathbf{s}}, \tilde{\mathbf{e}}$ follow a uniform distribution in $\{-k, k\}$. As a result, an adversary who can efficiently solve $LWE_{m,n,\sigma_k,q}$ can also deal with $LWE_{m,n,\sigma',q}$.

Besides, it is easy to check that there is a reduction from $LWE_{m,n,\sigma_k,q}$ to $LWE_{m,n,\sigma',q}$ when q is a prime by using a similar method. As a result, when q is a prime (as the case in this paper), the hardness of $LWE_{m,n,\sigma_k,q}$ equals to $LWE_{m,n,\sigma',q}$ which is known as binary-LWE and has been widely studied in [3, 8, 9, 13, 34, 39]. As a result, we will estimate the security of $LWE_{m,n,\sigma_k,q}$ by $LWE_{m,n,\sigma',q}$ in this paper.

Moreover, though the result will not be used in this paper, one can also prove that when q is a prime and s, e are separately sampled from arbitrary binary distributions (e.g. any $a \neq b \in \mathbb{Z}_q$), the hardness of corresponding LWE equals to binary-LWE with the same parameters.

2.5 Security Definition for Signature

In this section, we formally present precise definitions for the aforementioned security properties of digital signature schemes.

Definition 2.5 (BimodalSelfTargetMSIS_{H,n,q,k,l,β}). Suppose that $H : \{0, 1\}^* \times M \rightarrow \mathcal{R}_2$ is a cryptographic hash function. For positive integers q, k, l , a positive real number β and the dimension n of $\mathcal{R}$, we say that the advantage of an adversary $\mathcal{A}$ solving the search-BimodalSelfTargetMSIS_{H,n,q,k,l,β} problem with respect to $\mathbf{j} \in \mathcal{R}_2^k \setminus \{0\}$ is

$$\text{Adv}_{H,n,q,k,l,\beta}^{\text{BimodalSelfTargetMSIS}}(\mathcal{A}) = \Pr \left[\begin{array}{l} 0 < \|\mathbf{y}\|_2 < \beta \wedge \\ H(\mathbf{A}\mathbf{y} - qc\mathbf{j} \bmod 2q, M) = c \end{array} \middle| \begin{array}{l} (\mathbf{A}_0, \mathbf{b}) \leftarrow \mathcal{R}_q^{k \times (l-1)} \times \mathcal{R}_q^k; \\ \mathbf{A} = (-2\mathbf{b} + q\mathbf{j} | 2\mathbf{A}_0 | 2\mathbf{I}_k) \bmod 2q; \\ (\mathbf{y}, c, M) \leftarrow \mathcal{A}^{|H(\cdot)\rangle}(\mathbf{A}) \end{array} \right].$$

In the ROM (resp. QROM), the adversary is given classical (resp. quantum) access to H .

Theorem 2.1 (Classical Reduction from MSIS to BimodalSelfTargetMSIS). Assume that q is odd, $H : \{0, 1\}^* \times M \rightarrow \mathcal{R}_2$ is a cryptographic hash function modeled as a random oracle and that every polynomial time classical algorithm has a negligible advantage against MSIS_{H,n,q,k,l,β}. Then every polynomial-time classical algorithm has negligible advantage against BimodalSelfTargetMSIS_{H,n,q,k,l,β/2}.

Proof. Consider a BimodalSelfTargetMSIS_{H,n,q,k,l,β/2} classical algorithm $\mathcal{A}$ that is polynomial-time and has classical access to H . If $\mathcal{A}^{|H(\cdot)\rangle}(\mathbf{A})$ makes Q hash queries $H(\mathbf{w}_i, M_i)$ for $i = 1, \dots, Q$ and outputs a solution $(\mathbf{y}, c, M_j)$ for some $j \in [Q]$, then we can construct an adversary $\mathcal{A}'$ for MSIS_{n,q,k,l,β} as follows.

The adversary $\mathcal{A}'$ can first rewind $\mathcal{A}$ to the point at which the i -th query was made and reprogram the hash as $H(\mathbf{w}_i, M_i) = c' (\neq c)$. Then, with probability approximately $1/Q$, algorithm $\mathcal{A}$ will produce another solution $(\mathbf{y}', c', M_i)$. We then have

$$\mathbf{A}\mathbf{y} - qc\mathbf{j} = \mathbf{w}_i = \mathbf{A}\mathbf{y}' - qc'\mathbf{j} \bmod 2q \text{ and } \|\mathbf{y}\|_2, \|\mathbf{y}'\|_2 < \beta/2.$$

As q is odd, we have $\mathbf{A}(\mathbf{y} - \mathbf{y}') = (c - c')\mathbf{j} \bmod 2$. The fact that $c' \neq c$ implies that the latter is non-zero modulo 2, and hence so is $\mathbf{y} - \mathbf{y}'$ over the integers. As it also satisfies $(-\mathbf{b} | \mathbf{A}_0 | \mathbf{I}_k) \cdot (\mathbf{y} - \mathbf{y}') = \mathbf{0} \bmod q$ and $\|\mathbf{y} - \mathbf{y}'\|_2 < \beta$, it provides a MSIS_{n,q,k,l,β} solution for the matrix $(-\mathbf{b} | \mathbf{A}_0 | \mathbf{I}_k)$, where the submatrix $(-\mathbf{b} | \mathbf{A}_0) \in \mathcal{R}_q^{k \times l}$ is uniform. $\square$

The above classical reduction from MSIS to BimodalSelfTargetMSIS is the same with HAETAE. Moreover, since the reduction relies on the forking lemma,

it cannot be directly extended to a quantum reduction in the QROM. So we will then introduce the definition of Existential Unforgeability under No Message Attacks(EU-NMA) which will be used in our security proof.

Definition 2.6 (Existential Unforgeability against Non-adaptive Message Attacks(EU-NMA)). *In the EU-NMA, the adversary attempts to forge a signature without access to any valid message-signature pairs beforehand, relying solely on the public key for the attack. For a signature scheme $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$, the advantage of an EU-NMA adversary $\mathcal{A}$ is defined as:*

$$\text{Adv}_{\mathcal{S}}^{\text{EU-NMA}}(\mathcal{A}) = \Pr[\text{Verify}(vk, M, \sigma) = 1 \mid (sk, vk) \leftarrow \text{KeyGen}; (M, \sigma) \leftarrow \mathcal{A}(vk)].$$

Definition 2.7 (Existential Unforgeability under Chosen Message Attacks (EU-CMA)). *In EU-CMA, the adversary $\mathcal{A}$ can adaptively choose messages and obtain their corresponding signatures but cannot forge a valid signature for a new message that has never been signed before. Let $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ be a signature scheme and let (M_i, σ_i) be the set of message-signature pairs queried by the adversary and generated by the signer, for $i = 1, 2 \dots Q$. The adversary ultimately attempts to forge and outputs (M^*, σ^*) , while the advantage of $\mathcal{A}$ is defined as:*

$$\text{Adv}_{\mathcal{S}}^{\text{EU-CMA}}(\mathcal{A}) = \Pr \left[\begin{array}{l} M^* \notin \{M_i\}_{i \in [Q]} \wedge \\ \text{Verify}(vk, M^*, \sigma^*) = 1 \end{array} \middle| \begin{array}{l} (sk, vk) \leftarrow \text{KeyGen}; \\ (M^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sign}(sk, \cdot)} \end{array} \right].$$

Definition 2.8 (Strong Existential Unforgeability under Chosen Message Attacks(sEU-CMA)). *Compared to the EU-CMA, the sEU-CMA imposes stricter constraints on the adversary's forgery attempts: the forged signature-message pair must neither contain any previously queried messages nor replicate any previously obtained signatures. In other words, the adversary cannot succeed merely by finding a collision with an existing signature. Let $\mathcal{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ be a signature scheme. An sEU-CMA adversary $\mathcal{A}$ has access to the verification key and a signing oracle to make adaptive queries. Let the queried messages and the received signatures be (M_i, σ_i) for $i = 1, 2 \dots Q$. At the end of the experiment, it outputs a message-signature pair (M^*, σ^*) . Then the advantage of $\mathcal{A}$ is defined as:*

$$\text{Adv}_{\mathcal{S}}^{\text{sEU-CMA}}(\mathcal{A}) = \Pr \left[\begin{array}{l} (M^*, \sigma^*) \notin \{(M_i, \sigma_i)\}_{i \in [Q]} \\ \wedge \text{Verify}(vk, M^*, \sigma^*) = 1 \end{array} \middle| \begin{array}{l} (sk, vk) \leftarrow \text{KeyGen}; \\ (M^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sign}(sk, \cdot)} \end{array} \right].$$

2.6 Security Definition for Identification Protocol

To refine our reduction process, we define a set of properties for Fiat-Shamir identification protocols and analyze their roles in assisting the security proof of our signature scheme.

In the identification scheme, we require a sufficiently high probability for the honest entity to successfully pass the authentication.

Definition 2.9 (Identification Scheme). *An identification scheme is a tuple of PPT algorithms $ID = (Igen, P, V)$ such that:*

- *Igen: On input the security parameter 1^λ , algorithm Igen outputs a verification key vk and a signing key sk . We assume that vk defines the challenge space $\mathcal{C}$.*
- *P : The prover $P = (P_1, P_2)$ is split into two algorithms: given sk , algorithm P_1 produces a commitment w (first message sent to the verifier) and a state st ; algorithm P_2 , on input (sk, w, st) and a uniformly random challenge $c \in \mathcal{C}$ sent by the verifier in response to commitment w , outputs an answer z .*
- *V : On input (vk, w, c, z) , the deterministic verifier V outputs 1 or 0.*

Definition 2.10 (Completeness and Commitment-recoverability).

An identification scheme $ID = (Igen, P, V)$ is ϵ -complete for some $\epsilon > 0$ if for any $(vk, sk) \leftarrow Igen(1^\lambda)$, for any challenge $c \in \mathcal{C}$, we have:

$$\Pr[V(vk, w, c, z) = 0 \mid (w, c, z) \leftarrow (P(sk, vk) \leftrightarrow V(vk))] \leq \epsilon$$

where the randomness is taken over the random coins of P.

In addition, we need to ensure the commitment's recoverability, meaning that for the same public key, challenge and answer, at most one commitment can be generated. Also there exists a PPT algorithm that can recover the commitment using the answer and challenge such that $w = \text{Rec}(vk, c, z)$. In the signature scheme, the honest verifier rejects the signature if $\|z\|$ exceeds the predefined bounds. Thus, the completeness of our scheme solely depends on the boundary density estimation of z .

For an identification scheme ID and a hash function H (modeled as a random oracle in the analysis), we let $FS[ID, H]$ denote the resulting signature scheme.

We then recall the definitions of honest-verifier zero-knowledge and commitment min-entropy, which allow to reduce EUF-CMA security of $FS[ID, H]$ to its EUF-NMA security.

Definition 2.11 (HVZK and Commitment Min-entropy). *For an identification scheme $ID = (Igen, P, V)$, it is Honest-Verifier Zero-Knowledge if there exists a PPT simulator $\mathbf{Sim}$ such that, for any honest verifier V , the following distributions are computationally indistinguishable:*

- *Real transcription $(w, c, z) \leftarrow (P(sk, vk) \leftrightarrow V(vk))$.*
- *Simulated transcription $(w, c, z) \leftarrow \mathbf{Sim}(c, vk)$.*

An identification scheme satisfies α - Min Entropy or has α bits of commitment min-entropy if for any (vk, sk) in the range of I_{Gen} :

$$H_{\infty}(w \mid (w, c, z) \leftarrow (P(sk, vk) \leftrightarrow V(vk))) \geq \alpha,$$

where H_{∞} denotes information entropy.

Definition 2.12 (Lossy Identification Scheme). We say that an identification scheme $ID = (I_{\text{Gen}}, P, V)$ is lossy if it satisfies key-indistinguishability and ε_{ls} -lossy-soundness.

- key-indistinguishability: there exists a PPT lossy key generation algorithm LossyIGen that, on input a security parameter, outputs a verification key vk_{ls} , such that for any (possibly quantum) adversary $\mathcal{A}$

$$\text{Adv}^{\text{key-ind}}(\mathcal{A}) := |\Pr(\mathcal{A}(vk_{ls}) = 1) - \Pr(\mathcal{A}(vk) = 1)| = \text{negl}(\lambda),$$

where $vk_{ls} \leftarrow \text{LossyIGen}(1^\lambda)$ and $(vk, sk) \leftarrow I_{\text{Gen}}(1^\lambda)$ and the probability is taken over the random coins of $\mathcal{A}$ and LossyIGen and I_{Gen} .

- ε_{ls} -lossy-soundness for some $\varepsilon_{ls} > 0$: for any (unbounded) P^* interacting with V , we have:

$$\Pr[V(vk_{ls}, w, c, z) = 1 \mid (w, c, z) \leftarrow (P^*(vk_{ls}) \leftrightarrow V(vk_{ls}))] \leq \varepsilon_{ls}.$$

If we only consider classical adversaries, EU-NMA security of $\text{FS}[ID, H]$ can be argued by relying on the simpler notion of lossy special soundness.

Definition 2.13 (Lossy Special Soundness). An identification scheme is said to satisfy lossy special soundness if, given any two accepting transcripts (w, c, z) and (w, c', z') produced with the same commitment w but with distinct challenges $c \neq c'$, there exists an efficient extractor that can recover the unique secret witness associated with the commitment. This property must hold even when the public key is generated in a "lossy" mode.

An identification scheme has unique response if, for any (w, c) , there is at most one z such that the transcript (w, c, z) is accepted by the verifier. While this is not true by design for this scheme, we follow the proof idea of G+G by relaxing this property to computational unique response (CUR), which states that it is difficult to come up with two different responses for (w, c) .

Definition 2.14 (Computational Unique Responses). A Fiat-Shamir identification scheme is said to have computational unique responses if, for any fixed commitment w and c , there is at most one valid response z that will be accepted by the verifier-except with negligible probability:

$$\Pr[z_0 \neq z_1 \wedge V(vk, w, c, z_0) = V(vk, w, c, z_1) = 1 \mid (w, c, z_0, z_1) \leftarrow \mathcal{A}(vk)]$$

is $\text{negl}(\lambda)$, where the probability is over the choice of vk and the coins of $\mathcal{A}$.

3 Signature Scheme

3.1 Design Rationale

A brief recap on Fiat-Shamir Signature. Fiat-Shamir signature is a well-known signature structure, which is generalized from Schnorr’s identification protocol. The public key is a pair of matrices $(\mathbf{A}, \mathbf{T} = \mathbf{AS} \bmod q)$, where $\mathbf{A}$ is a uniform matrix modulo some integer q and $\mathbf{S}$ is a small-magnitude matrix that makes up the secret key. A signature for a message M is comprised of an integer vector $\mathbf{z}$ of the form $\mathbf{y} + \mathbf{S}c$, for some random small-magnitude $\mathbf{y}$ and some small-magnitude challenge $c = H(\mathbf{A}\mathbf{y} \bmod q, M)$. An important point of designing signature with Fiat-Shamir structure is to deal with the leakage of the secret information in $\mathbf{z} = \mathbf{y} + \mathbf{S}c$. For example, flooding, rejection sampling and Gaussian convolution theorem are several ways for solving this problem with different properties. At last, the verifier checks that the vector $\mathbf{z}$ is short and that $c = H(\mathbf{Az} - \mathbf{T}c \bmod q, M)$.

Bimodal distributions. The design of our signature follows the idea of G+G and HAETAE, which use bimodal distributions as a key stone. This technology was first introduced in [18]. More specially, in [18] the key generation process forces the equality $\mathbf{AS} = q\mathbf{I} \bmod 2q$, then the signature is now of the form $\mathbf{z} = \mathbf{y} + (-1)^b \mathbf{S}c$, where y is sampled from a fixed distribution and $b \in \{0, 1\}$ is sampled uniformly. To make sure that the verification test passes, computations are performed modulo $2q$ because $\mathbf{Az} = \mathbf{w} + qc = \mathbf{w} - qc \bmod 2q$. This modification can lead to more compact signatures than the unimodal setup. And based on this structure, many improvements have been proposed. For example, HAETAE uses uniform distributions over hyperballs to reduce the size of signatures instead of the discrete Gaussian distributions used in the original version. While G+G also uses the property of $\mathbf{AS} = q\mathbf{I} \bmod 2q$ to erase the influence of $\mathbf{h}$, since the support of $\mathbf{h}$ is $2\mathbf{SZ}^k$.

Compact signature size. As introduced above, lots of signatures are based on Fiat-Shamir structure, e.g. Dilithium and HAETAE which are known as the selected algorithms for NIST standardization and KPQC standardization respectively. For dealing with the leakage of the secret information in $\mathbf{z} = \mathbf{y} + \mathbf{cs}$, flooding, rejection sampling and Gaussian convolution theorem are several effective ways. In this paper, we introduce another way called 3C sampling. In 3C sampling, the output distribution is naturally independent of secret keys which results in a very compact signature, e.g. we can choose the module as $q = 257$. Besides, unlike Gaussian sampling or uniform hyperball sampling that are used in HAETAE and G+G, we only use integer in 3C sampling. This simplifies the implementation both in software and hardware. Moreover, we use the compres-

sion technique of encoding signatures, as it is suggested in [23] in the context of lattice-based hash-and-sign signatures, concerns the choice of the binary representation of the signature. As the largest part of the signature consists in a vector that follows the distribution made up by the convolution of several small distributions, we can choose some entropic coding to obtain a signature size close to its entropy. In particular, we choose the efficient range Asymmetric Numeral System to encode our signature, as it allows us to encode the whole signature and not lose a fraction of a bit per vector coordinate, like with Huffman coding.

Compact verification key. Instead of taking the challenge c as a vector over $\mathcal{R}$, we choose it in $\mathcal{R}$ similarly to HAETAE and G+G. The main condition on the challenge is that it has high min-entropy, which is already the case for binary vectors over $\mathcal{R}$. As a result, the secret $\mathbf{S}$ can be chosen as a vector over $\mathcal{R}$ rather than a matrix. The key-pair equation $\mathbf{AS} = q\mathbf{I}$ then becomes $\mathbf{As} = q\mathbf{j}$, where $\mathbf{j}$ is the vector starting with 1 and then continuing with 0. Besides, to store an element of $\mathbf{b}$ which is uniformly distributed in $q = 257$ needs 9 bits. However, the highest bit is 0 with high probability since $\mathbf{b}$ is statistically close to uniform distribution. By taking the probability into consideration, we can use a simplified huffman coding method to achieve smaller size of verification key. For an integer $x \in \mathbb{Z}_q$, $\text{EncodeToByte}(x) = \text{Bits}(x)$ if $0 \leq x < 255$ and $\text{EncodeToByte}(x) = \text{Bits}(255) \parallel \text{Bits}(x - 255)$ if $255 \leq x < 257$. In this way, most of elements of $\mathbb{Z}_q$ can be represented by 8 bits, and only 2 of out 257 elements need 9 bits. As a result, the average length for storing an element of $\mathbf{b}$ is $\frac{2}{257} \times 9 + \frac{255}{257} \times 8 = 8.008$ bits.

Algorithm 1 *EncodeToByte*

Input: $x \in \mathbb{Z}_q$;

Output: $x' \in \{0, 1\}^*$;

```

1:  $x' = 0$ 
2: if  $x < 255$  then
3:    $x' = \text{Bits}(x)$ 
4: else
5:    $x' = \text{Bits}(255) \parallel \text{Bits}(x - 255)$ 
6: end if
7: return  $x'$ 

```

Efficient choice of modulus. To realize a balance of efficiency and signature size, we need to choose the prime q to be a good prime in the sense that the ring operations can be implemented efficiently and suitable for security. For ring operations, we use the Number Theoretic Transform (NTT) with a fully split-

Algorithm 2 *DecodeFromByte*

Input: $x' \in \{0, 1\}^*$;**Output:** $x \in \mathbb{Z}_q$;

```

1:  $x = 0$ 
2: if  $x'_0 x'_1 \dots x'_7 = 0 \times \text{FF}$  then
3:    $x = 255 + x'_8$ 
4: else
5:    $x = \text{Integer}(x')$ 
6: end if
7: return  $x$ 

```

ting polynomial ring so q should be a small prime satisfying $n|(q-1)$, i.e. $q = 257$.

Security assumptions. Our schemes are based on the hardness of MLWE and MSIS problems, which have been widely studied and used in many designs including Fips 202 and 203. The difference is that we use binary-LWE settings where s, e are sampled uniformly from $\{0, 1\}^n$. Binary-LWE is used to design efficient schemes such as BLISS [18], GLP [25] and some FHE schemes [14, 20]. And its security has been studied in [3, 8, 9, 13, 34, 39]. Besides, Micciancio and Peikert also proved that LWE with binary error reduces to worst-case lattice problems when the number of samples is restricted to $n + O(n/\log(n))$ in [34]. In our designs, the number of samples is restricted to n thus ensures the security of binary-LWE.

3.2 Advantages and Limitations

Advantages

- Our schemes rely on the difficulty of hard lattice problems, MLWE and MSIS, which have been well-studied for a long time.
- The sum of signature and public key sizes are 59% to 63% smaller than those of Dilithium at comparable security levels.
- Implementation-wise, while our design rationale departs from HAETAE and G+G's, the scheme remains implementation-friendly. In particular, all integer computations means no need for floating-point or fixed-point arithmetic, which makes our efficiency similar to Dilithium.

Limitations

- No.

3.3 Key Strategy

3.3.1 Schnorr's Identification Protocol

Schnorr's identification protocol allows secure authentication between a prover and a verifier based on the hardness on the discrete logarithm problem in a cyclic group of order p , generated by an element g . The prover's public verification key is simply a group element g^s , whose discrete logarithm s forms the prover's signing key. The identification protocol proceeds as follows: the prover first commits to some uniform $y \leftarrow U(\mathbb{Z}_p)$ by sending g^y to a verifier. The latter returns some challenge $c \in \mathbb{Z}_p$, to which the prover replies with a response z , namely $z = y + cs \bmod p$. Here, no information about s is revealed as z is still uniform modulo p . However, a verifier is convinced that the prover knows s as it can verify $g^z = g^y(g^s)^c$. This can be compiled into a signature scheme by using the Fiat-Shamir heuristic.

When it comes to the case in lattice setting such as the schemes proposed in [29] and [30], now the private key is a tall matrix $\mathbf{S} \in \mathbb{Z}^{k \times m}$, and the public key is computed by $\mathbf{S}$ multiplying a uniform matrix $\mathbf{A} \in \mathbb{Z}_q^{m \times k}$, denoted as $\mathbf{T} = \mathbf{AS} \in \mathbb{Z}_q^{m \times m}$. Besides the masking vector $\mathbf{y} \in \mathbb{Z}^k$ is sampled according to a distribution and the commitment is computed as $\mathbf{w} = \mathbf{Ay}$. The challenge $\mathbf{c}$ belongs to a finite subset of $\mathbb{Z}^m$. Then the signature is computed as $\mathbf{z} = \mathbf{y} + \mathbf{Sc}$. The verifier is easy to check $\mathbf{Az} = \mathbf{w} + \mathbf{Tc}$. While the main problem is that the distribution of $\mathbf{y}$ should be set properly because a large bound leads to forge attacks and a small bound may reveal the information of $\mathbf{s}$.

One way to deal with the problem is to choose a large standard deviation of $\mathbf{y}$ to flood the center, which leads to very large signature size. Another way is using rejection sampling to erase the information of $\mathbf{S}$ such as HAETAE and Dilithium, this method needs to repeat several times to perform rejection sampling. Moreover, a side channel attack may make use of the rejection sampling according to [41].

Besides, G+G proposed the third way by choosing the distribution of $\mathbf{y}$ according to $\mathbf{S}$, sampling another variant $\mathbf{h}$ according to $\mathbf{c}$, which ensures $\mathbf{z} = \mathbf{y} + \mathbf{Sc} + \mathbf{h}$ is close to a known distribution χ_z . They use the convolution theorem of Gaussian distribution to control the distance between $\mathbf{z}$ and χ_z , but this method uses high dimensional Gaussian sampling which can not be easily implemented.

3.3.2 Convolution of Constant Counts Sampling

Our scheme follows the idea of G+G but using a new sampling method rather than Gaussian convolution theorem to prevent the leakage of the secret key information from $\mathbf{z}$. In this subsection, we show how to choose the distributions of $\mathbf{y}, \mathbf{h}$ in order to cover the information of $\mathbf{S}$ in $\mathbf{z} = \mathbf{y} + \mathbf{Sc} + \mathbf{h}$ in our scheme. Before we describe the details of our sampling method, we first give a definition

of “opposite” distribution and then note the following two observations.

“Opposite” distribution. Let Pr_X be any probability distribution defined on the measurable space $(\mathbb{R}, \mathfrak{B})$, where $\mathfrak{B}$ is the Borel σ -algebra. Define its “Opposite” distribution, denoted as $Y = -X$, whose probability measure Pr_Y satisfies: For any Borel set $A \in \mathfrak{B}$, $Pr_Y(-A) = Pr_X(A)$, where $-A = \{-x | x \in A\}$. In other words, if the probability of the event $Y \in -A$ always equals the probability of the event $X \in A$, we write $X = -Y$ or $Y = -X$.

Observation 1. For any symmetric distribution ψ with the support $\mathbb{Z}$ and the center 0, we always have $\psi = -\psi$.

Observation 2. Let $\Psi(n, \mathbf{b}) = \sum_{i=0}^n (-1)^{b_i} \psi_i$ where $b_i \in \{0, 1\}$ and all ψ_i are independent symmetric distributions with the support $\mathbb{Z}$ and the center 0. Then based on the observation 1, we have $\Psi(n, \mathbf{b}_1) = \Psi(n, \mathbf{b}_2)$ for any $\mathbf{b}_1, \mathbf{b}_2 \in \{0, 1\}^n$.

Secondly, we define the basic distribution. Denote the distribution of uniformly choosing from $\{-k, k\}$ as σ_k . Let $\chi(c, k)$ denote a distribution with center c and support $k\mathbb{Z}$, we define $\chi(0, 1), \chi(0, 2)$ and $\chi(-1, 2)$ as follows which will be used in our sampling process. And their sampling algorithm can be found in algorithm *Sample- χ* .

$$Pr(\chi(0, 1)) = \begin{cases} \frac{1}{4} & \text{output } -1, \\ \frac{1}{2} & \text{output } 0, \\ \frac{1}{4} & \text{output } 1. \end{cases}$$

$$Pr(\chi(0, 2)) = \begin{cases} \frac{1}{4} & \text{output } -2, \\ \frac{1}{2} & \text{output } 0, \\ \frac{1}{4} & \text{output } 2. \end{cases}$$

$$Pr(\chi(-1, 2)) = \begin{cases} \frac{1}{2} & \text{output } -2, \\ \frac{1}{2} & \text{output } 0. \end{cases}$$

Thirdly, we show how to choose the distribution of $\mathbf{h}$ and $\mathbf{y}$. As for $\mathbf{h}$, it is necessary to use $\mathbf{h}$ to erase the center of $\mathbf{S}\mathbf{c}$ and make sure $\mathbf{A}\mathbf{h} = \mathbf{0} \bmod 2q$. Since $\mathbf{A}\mathbf{S} = q\mathbf{I} \bmod 2q$, the support of $\mathbf{h}$ must be $2\mathbf{S}\mathbb{Z}^m$. We can set $\mathbf{h} = \mathbf{S}\mathbf{x}$ where $x_i \leftarrow \chi(-1, 2)$ for $c_i = 1$ and $x_i = 0$ for $c_i = 0$. It can be checked that the distribution of $x_i + c_i$, denoted as $\xi(c_i)$, satisfies $\xi(c_i) = x_i + c_i = \sigma_1$ when $c_i = 1$ and $\xi(c_i) = x_i + c_i = 0$ when $c_i = 0$, thus the distribution of $\xi(c_i) = x_i + c_i$ is always symmetric with the center 0. As a result, we have $\gamma = \mathbf{S}\mathbf{c} + \mathbf{h} = \mathbf{S}(\mathbf{c} + \mathbf{x}) \in \mathbb{Z}_q^k$, and $\gamma_i = \sum_{j=1}^m S_{i,j}(x_j + c_j)$.

Suppose all elements in $\mathbf{S}$ are chosen from $\{-1, 1\}$ (or we can view $\mathbf{S}_i$ are sampled from a special constant hamming sampling with $\tau_s = k$) and $\mathbf{c}$ is a vector with an even fixed hamming weight τ .

Then according to the observation 2, we have $\gamma_i = \sum_{j=1}^m S_{i,j}(x_j + c_j)$ follows the distribution of $\sum_{j=1}^m \xi(c_i)$ since $\xi(c_i) = x_i + c_i$ is always symmetric with the center 0. As the hamming weight of $\mathbf{c}$ is τ , we have $\gamma_i \leftarrow \sum_{j=1}^{\tau} \sigma_1$. And by observing $\sigma_1 + \sigma_1 = \chi(0, 2)$, we have $\gamma_i \leftarrow \sum_{j=1}^{\tau/2} \chi(0, 2)$ which have the support $2\mathbb{Z}$. Now that γ_i naturally covers the information of $\mathbf{S}$ since it always has the same distribution for any $\mathbf{S} \in \{-1, 1\}^{k \times m}$.

Then we will show how to choose the distribution of $\mathbf{y}$ to provide enough min-entropy and also avoid ratio attack. The distribution of $\mathbf{y}$ consists of two parts. The first part is sampled independently to provide entropy, and the second part is to sample several variables that will be shared through each elements of $\mathbf{y}$ to avoid ratio attack.

Firstly we discuss the first part of $\mathbf{y}$, denoted as $\mathbf{y}_1$. Since γ_i have the support $2\mathbb{Z}$ and $z_i = y_i + \gamma_i$, it is seen that z_i and y_i share the same parity. So we need to choose the distribution of $\mathbf{y}_1$ to ensure that it have large enough entropy by giving its parity, i.e. we can sample $y_{1,i}$ from $\chi(0, 1) + \chi(0, 1)$. At this time, the probabilities of $y_{1,i}$ follows: $Pr(y_{1,i} = 2) = Pr(y_{1,i} = -2) = \frac{1}{16}$, $Pr(y_{1,i} = 1) = Pr(y_{1,i} = -1) = \frac{1}{4}$, $Pr(y_{1,i} = 0) = \frac{3}{8}$. As a result, the conditional entropy of y_1 can be computed as: $H(y_{1,i}|y_{1,i} \bmod 2) = -Pr(y_{1,i} \bmod 2 = 0) \cdot \sum_{y_{1,i}} Pr(y_{1,i}|y_{1,i} \bmod 2 = 0) \log_2(Pr(y_{1,i}|y_{1,i} \bmod 2 = 0)) - Pr(y_{1,i} \bmod 2 = 1) \cdot \sum_{y_{1,i}} Pr(y_{1,i}|y_{1,i} \bmod 2 = 1) \log_2(Pr(y_{1,i}|y_{1,i} \bmod 2 = 1)) = 1.03bit$. As elements of $\mathbf{y}_1$ are independent of each others, the entropy of $\mathbf{y}_1$ is $1.03k$.

Then we discuss the second part of $\mathbf{y}$, denoted as $\mathbf{y}_2$. The main reason of sampling $\mathbf{y}_2$ is to provide security against ratio attack which is proposed in [40] to attack G+G. The main idea of ratio attack is to utilize the truth that $s_0 = 1 \in R_q$ which is different from other parts of $\mathbf{S}$. So the distribution of $z_0 = y_0 + c$ are also different from other parts of $\mathbf{z}$. Besides, $z_{0,0}, z_{i,j}$ share the same masking element $\gamma_0 = x_0 + c_0$, this makes $\frac{z_{i,j}}{z_{0,0}}$ a Cauchy distribution who is related to $s_{i,j}$ and γ_0 since other γ_i and y_i are all independent of each other. As a result, an attacker can reveal the information of $\mathbf{S}$ by collecting numbers of signatures and compute the expectation of $z_{i,j}/z_{0,0}$. To avoid this attack, we introduce more relationships between $z_{i,j}$ and $z_{0,0}$ through $\mathbf{y}_2$.

To sample $\mathbf{y}_2$, we first generate a random challenge vector c' with hamming weight τ and compute its corresponding x' such that $\xi(c'_i) = x' + c'$ ensures that $\xi(c'_i)$ follows the same distribution of $\xi(c_i)$. Then we randomly sample an element $\bar{s}'_0 \in \mathcal{R}_q$ which follows the same distribution of the element in $\mathbf{s}_{\text{gen}}$, and sample a vector $\mathbf{s}' = (s'_0, s'_1, \dots, s'_{k+l-1}) \in \mathcal{R}_q^{k+l}$ with every s'_i follows the same distribution of the element in $\mathbf{s}_{\text{gen}}$. At last, we compute $\mathbf{y}_2 = (x' + c') \cdot (\bar{s}'_0 + s'_0, s'_1, \dots, s'_{k+l-1})$, and we finally have $\mathbf{y} = \mathbf{y}_1 + \mathbf{y}_2$.

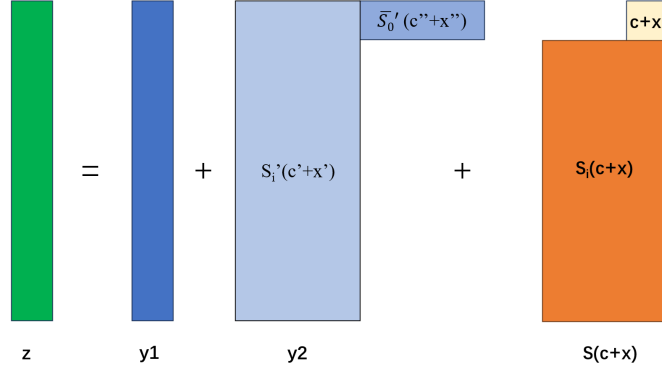


Figure 1. 3C Sampling Method.

As a result, now we can define the distribution of $\mathbf{z}$. Each element of $\mathbf{z}$, denoted as $z_{i,j}$, follows the same distribution which is the convolution of a fixed number of $\chi(0, 2)$ and two $\chi(0, 1)$ (except for the first element of $\mathbf{z}$, which has an extra $\xi(c_i)$). Since all these distributions are independent with the secret key, we can check the distribution of $\mathbf{z} = \mathbf{y} + \mathbf{S}\mathbf{c} + \mathbf{h}$ can be public sampled for any $\mathbf{S}, \mathbf{c}$ under the proposed settings.

It is very interesting to see that γ_i naturally cover the information of $\mathbf{S}$ due to the symmetry of the distributions. In this way, we don't need to limit the variance of $\mathbf{y}$ to satisfy convolution theorem or reduce the statistical distance between target and source distributions. In this way, a compact signature can be obtained. Besides, this property can also be used to effectively compress the signature size by using encoding method because z_i is a distribution with its probabilities publicly known. And we call this method as Convolution of Constant Counts Sampling(3C sampling).

Algorithm 3 *Sample- χ*

Input: $c \in \{-1, 0\}, k > 0, rt$;

Output: $x \in \mathbb{Z}$;

```

1:  $x = 0$ 
2: if  $c = 0$  then
3:    $rd = rt.getbit(2)$ 
4:    $x = k \times (rd[0] - rd[1])$ 
5: else
6:    $rd = rt.getbit(1)$ 
7:    $x = rd \times (-2)$ 
8: end if
9: return  $x$ 

```

Algorithm 4 *Sample_{-X}*

Input: $c \in \{-1, 0\}, rt$;**Output:** $x \in \mathbb{Z}$;

```

1:  $x = 0$ 
2: if  $c = 0$  then
3:    $x = 0$ 
4: else
5:    $x = \text{Sample}_{-\chi}(c, 2, rt)$ 
6: end if
7: return  $x$ 

```

Algorithm 5 *Sample_{-Y₁}*

Input: rt ;**Output:** $y_1 \in \mathbb{Z}$;

```

1:  $y_1 = \text{Sample}_{-\chi}(0, 1, rt) + \text{Sample}_{-\chi}(0, 1, rt + 2)$ 
2: return  $y_1$ 

```

3.3.3 Challenge Sampling

As mentioned in the previous subsection, we will take the challenge c as an element in $\mathcal{R}$. The challenges we use are polynomials $c \in \mathcal{R}$ with binary coefficients and the number of 1 is fixed as τ . The challenge space has size $\binom{n}{\tau}$, to sample such challenges we rely on the (binary version of) `SampleInBall` algorithm from Dilithium with slight modification because now $c_i \in \{0, 1\}$ instead of $c_i \in \{-1, 0, 1\}$.

3.3.4 Signature Encoding

Instead of splitting signature components into low and high bits to use hints, we directly encode the entire signature with a suitable encoding since our signature has a quite small public key. Arithmetic coding encodes the entire vector as a single number, while Huffman coding processes each coordinate independently. Unlike Huffman coding, arithmetic coding can approaches the theoretical entropy limit also for alphabets, where symbol probabilities are not powers of two. We adopt the recent range Asymmetric Numeral Systems (rANS) [21] entropy coding, which encodes states to natural numbers for efficient hardware implementation. The rANS supports stream processing and enables finite-precision integer arithmetic via renormalization, reducing hardware implementation complexity. In addition, it can be transformed into a lookup table-based implementation (tANS), eliminating arithmetic operations entirely to accelerate computation.

Definition 5 (Range Asymmetric Numeral System). Let $S \subseteq [0, 2^n - 1]$, $n \in \mathbb{N}$ and we define a function $g : S \rightarrow \mathbb{Z} \cap (0, 2^n]$, satisfying $\sum_{s \in S} g(s) \leq 2^n$ and $g(s) = 0$ for all $s \notin S$.

Then, We define the following: 1) $\text{CDF} : S \rightarrow \mathbb{Z}$, defined as $\text{CDF}(s) = \sum_{i=0}^{s-1} g(i)$ with $\text{CDF}(0) = 0$. 2) $\text{symbol} : \mathbb{Z} \rightarrow S$, defined as $s \in S$ satisfying $\text{symbol}(x) = s \iff \text{CDF}(s) \leq x < \text{CDF}(s+1)$. Finally, the function C is defined as:

$$C(x, s) = \left\lfloor \frac{x}{g(s)} \right\rfloor \cdot 2^n + (x \bmod^+ g(s)) + \text{CDF}(s).$$

The rANS encoding and decoding functions for the set S and frequency $g/2^n$ are defines as follows:

Algorithm 6 rANSEncode

Input: $\mathbf{z} = (z_1, \dots, z_m)$;
Output: x ;
1: $x = 0$
2: **for** $i = m$ **downto** 1 **do**
3: $s = \mathbf{z}[i]$
4: $x = \left\lfloor \frac{x}{g(s)} \right\rfloor \cdot 2^n + (x \bmod^+ g(s)) + \text{CDF}(s)$
5: **end for**
6: **return** x

Algorithm 7 rANSDecode

Input: x, m ;
Output: $S = (t_1, \dots, t_m)$;
1: $y = x$
2: $M = 2^n$
3: **for** $i = 1, \dots, m$ **do**
4: $s = y \bmod^+ M$
5: $t_i = \text{symbol}(s)$
6: $S[i] = t_i$
7: $y = g(t_i) \cdot \left\lfloor \frac{y}{M} \right\rfloor + s - \text{CDF}(t_i)$
8: **end for**
9: **return** S

Lemma 3.1. *The rANS coding enables lossless compression, where any sequence s can be perfectly reconstructed after encoding and decoding. The encoded*

size of rANS asymptotically equals the Shannon entropy of symbols, indicating that rANS coding approaches optimal compression efficiency. For any positive x and any probability distribution q over S , the following holds

$$\sum_{s \in S} q(s) \log(C(x, s)) \leq \log(x) + \sum_{s \in S} q(s) \log\left(\frac{g(s)}{2^n}\right) + \frac{2^n}{x}.$$

Finally, the cost of encoding the first symbol is bounded by n . Specifically, for any $x \in S$, the logarithmic value of the state transition function $C(0, s)$ satisfies $\log(C(0, s)) \leq n$, confirming the upper bound of the encoding complexity for the initial symbol.

In our signature scheme, the generation of signature $\mathbf{z}$ essentially involves superimposing multiple distributions. By designing the distribution characteristics of each component, we obtain the distribution of the final $\mathbf{z}$. The final result realizes the unification of distributions through convolution, enabling deterministic probability calculations to derive g .

3.4 Signature Algorithms

Key generation. Our key generation method follows a classical framework of bimodal distributions where we need to ensure $\mathbf{A}\mathbf{s} \bmod 2q = q\mathbf{j}$ as the case in HAETAE and G+G. And the difference is that every component of the secret is sampled uniformly from $\{-1, 1\}$.

To build a key pair $(\mathbf{A}, \mathbf{s})$ satisfying our requirements, we start by generating an MLWE sample $\mathbf{b} = \mathbf{A}_{\text{gen}}\mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \bmod \mathbf{q}$, where $\mathbf{A}_{\text{gen}} \leftarrow U(\mathcal{R}_q^{k \times (l-1)})$ and $(\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) \leftarrow (\sigma_1^{(l-1)} \times \sigma_1^k)$.

Then we define $\mathbf{A} = (-2\mathbf{b} + q\mathbf{j} | 2\mathbf{A}_{\text{gen}} | 2\mathbf{I}_k)$. And $\mathbf{s} = (1 | \mathbf{s}_{\text{gen}} | \mathbf{e}_{\text{gen}})$ which ensures that $\mathbf{A}\mathbf{s} = q\mathbf{j} \bmod 2q$. In practice, the verification key is then combination of $\mathbf{b}$ and the seed that allows generating $\mathbf{A}_{\text{gen}}$.

Sign process. Our sign process is similar to that in G+G where 3C sampling is utilized instead of Gaussian sampling.

In signature, it is important to evaluate the bound of $\mathbf{z}$. In our settings, the distribution of z_i denoted as $\chi_{z_i} = \sum^\tau \chi(0, 2) + \sum^2 \chi(0, 1)$ (except for χ_0 who has an extra $\xi(c)$). So the upper bound of $|z_i|$ is $2\tau + 2 + 1$ and we can evaluate it by computing the distribution of $\sum^{\tau+2} \chi(0, 2)$ for simplicity. The probability of $|z_i| \geq B$ can be estimated as:

$$Pr(|z_i| \geq B) \approx 2 \sum_{\delta=B/2}^{\tau+2} \sum_{\substack{m_1=m_2+\delta, \\ m_1+m_2+m_3=\tau+2, \\ m_1, m_2, m_3 \geq 0}} \binom{\tau+2}{m_1} \binom{\tau+2-m_1}{m_2} 2^{-2m_1-2m_2-m_3},$$

Algorithm 8 SIG_KeyGen**Input:** 1^λ ;**Output:** sk, pk ;

```

1:  $seed \leftarrow \{0, 1\}^{\rho_0}$ 
2:  $(seed_A, seed_{sk}, seed_{s'}, K) = H_{gen}(seed)$ 
3:  $\mathbf{A}_{gen} \in \mathcal{R}_q^{k \times (l-1)} := expandA(seed_A)$ 
4:  $(\mathbf{s}_{gen}, \mathbf{e}_{gen}) \in (\mathcal{R}_q^{l-1}, \mathcal{R}_q^k) := expandS(\sigma_1, seed_{sk})$ 
5:  $\mathbf{b} = \mathbf{A}_{gen} \cdot \mathbf{s}_{gen} + \mathbf{e}_{gen} \bmod q \in \mathcal{R}_q^k$ 
6:  $\mathbf{A} = (-2\mathbf{b} + q\mathbf{j}|2\mathbf{A}_{gen}|2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$ 
7:  $\mathbf{s} = (1, \mathbf{s}_{gen}, \mathbf{e}_{gen})$ 
8:  $\mathbf{b}' = EncodeToByte(\mathbf{b})$ 
9:  $\{\bar{s}'_0, s'_0, \dots, s'_{k+l-1}\} \in \mathcal{R}_q^{k+l+1} := expandS(\sigma_1, seed_{s'})$ 
10:  $\mathbf{s}' = \{s'_0, s'_1, \dots, s'_{k+l-1}\} \in \mathcal{R}_q^{k+l}$ 
11:  $\bar{\mathbf{s}}' = \{\bar{s}'_0, 0, \dots, 0\} \in \mathcal{R}_q^{k+l}$ 
12: return  $sk = (\mathbf{s}, \mathbf{s}', \bar{\mathbf{s}}', K), pk = (seed_A, \mathbf{b}')$ 

```

$$Pr(|\mathbf{z}|_\infty < B) = \prod_{i=0}^{(k+l)n-1} (1 - Pr(|z_i| \geq B)).$$

For ring settings, we take $n = 256$ through the whole article, so $Pr(|z_i| \geq B)$ and $Pr(|\mathbf{z}|_\infty < B)$ for the parameters of our scheme are shown below.

Table 1. Bound of z_i for $q = 257$

Security	128	192	256
(k, l)	(2, 3)	(3, 4)	(4, 5)
B	32	48	60
τ	30	60	128
challenge entropy $[\log(\frac{256}{\tau})]$	129	197	251
$Pr(z_i \geq B)$	$2^{-13.7}$	$2^{-15.6}$	$2^{-12.0}$
$Pr(\mathbf{z} _\infty < B)$	0.91	0.95	0.58

Verify process. Our verify process is similar to HAETAE and G+G without using hints. However, the computation of $\mathbf{Az}$ can be simplified in implementation. Since the last parts of $\mathbf{A}$ are $2\mathbf{Id}_k$, let $\mathbf{A} = (\mathbf{A}_0|2\mathbf{Id}_k)$, $\mathbf{z}^T = (\mathbf{z}_1^T, \mathbf{z}_2^T) \in \mathcal{R}_{2q}^l \times \mathcal{R}_{2q}^k$. Then we can compute $\mathbf{Az} = \mathbf{A}_0\mathbf{z}_1 + \mathbf{z}_2 + \mathbf{z}_2$ to decrease the number of multiplications for better efficiency.

3.5 Specification

We give the description of our signature scheme with the following building blocks:

Algorithm 9 SIG-Sign**Input:** sk, pk, M ;**Output:** σ ;

```

1:  $\mathbf{A}_{\text{gen}} \in \mathcal{R}_q^{k \times (l-1)} := \text{expandA}(\text{seed}_A)$ 
2:  $\mathbf{b} = \text{DecodeFromByte}(\mathbf{b}')$ 
3:  $\mathbf{A} = (-2\mathbf{b} + q\mathbf{j}|2\mathbf{A}_{\text{gen}}|2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$ 
4:  $\mu = H_{\text{gen}}(\text{seed}_A, \mathbf{b}', M)$ 
5:  $(\text{seed}_{y_{cx'}}, \text{seed}_x) = H_{\text{gen}}(K, \mu)$ 
6:  $\text{count} = 0$ 
7:  $(\text{seed}_y, \text{seed}_{c'}, \text{seed}_{x'}) = H_{\text{gen}}(\text{seed}_{y_{cx'}}, \text{count})$ 
8:  $c', c'' \in \mathcal{R}_q = \text{SampleInBall}(\text{seed}_{c'})$ 
9:  $x' \leftarrow \text{Sample}_X(-c', \text{expand}(\text{seed}_{x'})) \in \mathcal{R}_q$ 
10:  $x'' \leftarrow \text{Sample}_X(-c'', \text{expand}(\text{seed}_{x'})) \in \mathcal{R}_q$ 
11:  $\mathbf{y} \leftarrow (c' + x') \cdot \mathbf{s}' + (c'' + x'') \cdot \mathbf{s}' + \text{Sample}_Y_1(\text{expand}(\text{seed}_y))$ 
12:  $\mathbf{w} = \mathbf{A}\mathbf{y} \bmod 2q$ 
13:  $c \in \mathcal{R}_q = \text{SampleInBall}(\mathbf{w}, \mu)$ 
14:  $\text{seed}_x = H_{\text{gen}}(\text{seed}_x, \text{count})$ 
15:  $x \leftarrow \text{Sample}_X(-c, \text{expand}(\text{seed}_x)) \in \mathcal{R}_q$ 
16:  $\mathbf{z} = \mathbf{y} + c \cdot \mathbf{s} + x \cdot \mathbf{s}$ 
17: if  $\|\mathbf{z}\|_\infty \geq B$  then
18:    $\text{count}++$ 
19:   goto 7
20: end if
21:  $\mathbf{z}' = \text{Encode}(\mathbf{z})$ 
22: return  $(\mathbf{z}', c)$ 

```

Algorithm 10 SIG-Verify**Input:** $pk, M, \mathbf{z}', c$;**Output:** $\{0, 1\}$;

```

1:  $\mathbf{z} = \text{Decode}(\mathbf{z}')$ 
2:  $\mathbf{A}_{\text{gen}} \in \mathcal{R}_q^{k \times (l-1)} := \text{expandA}(\text{seed}_A)$ 
3:  $\mathbf{b} = \text{DecodeFromByte}(\mathbf{b}')$ 
4:  $\mathbf{A} = (-2\mathbf{b} + q\mathbf{j}|2\mathbf{A}_{\text{gen}}|2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_{2q}^{k \times (k+l)}$ 
5:  $\mathbf{w} = \mathbf{A}\mathbf{z} - qc \cdot \mathbf{j} \bmod 2q$ 
6:  $\mu = H_{\text{gen}}(\text{seed}_A, \mathbf{b}', M)$ 
7: if  $c = \text{SampleInBall}(\mathbf{w}, \mu)$  and  $\|\mathbf{z}\|_\infty < B$  then
8:   return 1
9: else
10:  return 0
11: end if

```

- Hash function H_{gen} for generating the seeds and hashing the messages,
- Extendable output function expandA for deriving $\mathbf{A}_{\text{gen}}$ from seed_A ,

- Extendable output function expandS for deriving $\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}$ from seed_{sk} and $\mathbf{s}'$ from $\text{seed}_{s'}$,
- Extendable output function expand for deriving a uniformly random bit string from seed ,

The above building blocks can be implemented with symmetric primitives. Specifically, we use SHAKE256 for the hash functions and Extendable output functions except for expandA .

3.6 Security

Existential Unforgeability under Chosen Message Attacks (EU-CMA) is regarded as a standard security notion for digital signature schemes. The adversary is given the verification key and has access to a signing oracle to call on (adaptively) chosen messages. The adversary wins if it forges a valid signature of a new, non-queried message. Strong Existential Unforgeability under Chosen Message Attacks (sEU-CMA) is a slightly stronger security notion than EU-CMA: the adversary wins if it forges a valid signature-message pair that it did not already see.

The security of our signature scheme follows that in HAETAE and G+G, where the reduction can be proved in two ways. The first way is performed in the classical Random Oracle Model (ROM) under the standard MLWE and MSIS assumptions.

And the second way is using the zero-knowledge property of the underlying identification scheme, Existential Unforgeability under No Message Attacks (EU-NMA) reduces to (s)EU-CMA security, and can be proved both in the ROM and the QROM, which is introduced in [15] and [4].

UF-NMA is related to a problem defined in HAETAE, denoted as Bimodal-SelfTargetMSIS. Note that in the ROM, MSIS reduces to BimodalSelfTargetMSIS, but the reduction is not tight and does not readily extend to quantum adversaries (it relies on the forking lemma). This said, this non-tightness and limitation to classical adversaries is not known to reflect any weakness. For setting parameters, we consider the hardness of MSIS and MLWE for relevant parameters. Intuitively, the MLWE assumption is used for security against key-recovery attacks, and the BimodalSelfTargetMSIS used for security against forgeries is identified to the MSIS assumption.

In this section, we first describe the Rhyme identification protocol, denoted as Rhyme_ID . Then in the next subsection, we prove the required properties to compile it into a signature using the Fiat-Shamir heuristic.

3.6.1 Description of the Rhyme Identification Protocol

Let us first introduce the parameters of the scheme as well as some notations.

Let $k, l > 0$ and $\mathbf{j} = \{1, 0, 0, 0 \dots\} \in \mathcal{R}^k$. Let σ and χ be distributions previously described. Let $c \in \mathcal{R}_2$ with hamming weight τ be the challenge space, which we assume to be finite. The scheme is also parametrized by an odd modulus q and an acceptance bound B .

The Rhyme identification protocol is described in Table 2. The instance generation algorithm samples a verification key $\mathbf{A} \in \mathbb{Z}_{2q}^{k \times (k+l)}$ and a signing key $\mathbf{S} \in \mathbb{Z}^{(k+l) \times 1}$ with small-magnitude coefficients such that $\mathbf{A} \cdot \mathbf{S} = q\mathbf{j} \bmod 2q$. In the first phase of the interaction, the prover samples a vector $\mathbf{y}$ from $\chi(0, 1) + \chi(0, 1) + \sum^{\tau/2} \chi(0, 2)$ (except for y_0 who is sampled from $\chi(0, 1) + \chi(0, 1) + \sum^{\tau} \chi(0, 2)$), and sends the commitment $\mathbf{w} = \mathbf{A}\mathbf{y} \bmod 2q$ to the verifier. The protocol is public-coin, i.e., the verifier just samples c uniformly in the challenge space and sends it to the prover. After receiving c , the prover samples a carefully designed vector $\mathbf{x}$. The distributions of $\mathbf{y}$ and $\mathbf{x}$ are set so that the final target distribution converges to a symmetric, publicly sampleable distribution.

IGen(1^λ)

1. $\mathbf{A}_{\text{gen}} \leftarrow U(\mathcal{R}_q^{k \times (l-1)})$
2. $(\mathbf{s}_{\text{gen}}, \mathbf{e}_{\text{gen}}) \leftarrow U(\sigma_1^{l-1}, \sigma_1^k)$
3. $\mathbf{b} \leftarrow \mathbf{A}_{\text{gen}} \cdot \mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}} \bmod q$
4. $\mathbf{A} \leftarrow (-2\mathbf{b} + q\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{Id}_k) \bmod 2q \in \mathcal{R}_q^{k \times (k+l)}$
5. $\mathbf{s} \leftarrow (1 \mid \mathbf{s}_{\text{gen}} \mid \mathbf{e}_{\text{gen}})$
6. $vk \leftarrow \mathbf{A}, sk \leftarrow \mathbf{s}$
7. **return** (vk, sk)

P(A,S)		V(A)
$y_i \leftarrow \chi(0, 1) + \chi(0, 1) + \sum^{\tau/2} \chi(0, 2)$		
$\mathbf{w} \leftarrow \mathbf{A}\mathbf{y} \bmod 2q$	$\xrightarrow{w}$	
	$\xleftarrow{c}$	$c \leftarrow U(\mathcal{C})$
$x \leftarrow \text{Sample}_X(c)$		
$\mathbf{z} \leftarrow \mathbf{y} + \mathbf{s}(c + x)$	$\xrightarrow{z}$	Accept if
		$\mathbf{Az} = \mathbf{w} + qc\mathbf{j} \bmod 2q$
		and $\ \mathbf{z}\ _\infty < B$

Table 2. The Rhyme Identification Protocol

By integrating this identification protocol with the Fiat-Shamir (without aborts) paradigm, we derive a lattice-based signature scheme denoted as FS[Rhyme_ID, H], as formalized in the following theorem. The correctness and security of the scheme directly follow from the properties of the underlying identification protocol.

3.6.2 Security proof

We progressively refine the security proofs of our cryptographic components required in the security reduction, which is tight as described in [28]. First, we should ensure our signature protocol is complete and commitment recoverable. As a corollary, we obtain that the resulting Fiat-Shamir signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is correct.

Theorem 3.1. *Let $k, l > 0$ in the range of IGen . Let B and ε_c be such that $\Pr_z[\|z\| \geq B] \leq \varepsilon_c$. Let $q > 2B$ be an odd modulus. Then the *Rhyme_ID* identification protocol is ε_c -complete and achieves commitment-recoverability.*

Proof. First, we note that $\mathbf{A}\mathbf{S} = \mathbf{q}\mathbf{j} \bmod 2q$ holds for any matrix pair output by IGen . Then, in order to pass the first verification step, a transcript $(\mathbf{w}, c, \mathbf{z})$ must satisfy:

$$\mathbf{A}\mathbf{z} = \mathbf{A}(\mathbf{y} + \mathbf{S}c + \mathbf{S}x) = \mathbf{w} + qc\mathbf{j} + \mathbf{0} \bmod 2q.$$

In particular, this defines a unique commitment $\mathbf{w} = \mathbf{A}\mathbf{z} - qc\mathbf{j} \bmod 2q$ such that $(\mathbf{w}, c, \mathbf{z})$ can be a valid transcript, and $\mathbf{w}$ is efficiently recoverable, by defining Rec as $\text{Rec} := \mathbf{A}\mathbf{z} - qc\mathbf{j} \bmod 2q$.

As for completeness, unlike the G+G scheme, where the convolved real distribution deviates from the expected spherical Gaussian, our sampling method achieves a distribution that can be perfectly sampled publicly (as described in Chapter 3.3.2). The resulting distribution of $\mathbf{z}$ is identical to the publicly specified distribution in the protocol, hence the probability of an honest transcript $(\mathbf{w}, c, \mathbf{z})$ failing the protocol is equivalent to the marginal density of the $\mathbf{z}$ distribution. Based on the previous discussion, we can conclude:

$$\begin{aligned} \Pr(|z_i| \geq B) &\approx \\ 2 \sum_{\delta=B/2}^{\tau+2} \sum_{\substack{m_1=m_2+\delta, \\ m_1+m_2+m_3=\tau+2, \\ m_1, m_2, m_3 \geq 0}} \binom{\tau+2}{m_1} \binom{\tau+2-m_1}{m_2} 2^{-2m_1-2m_2-m_3}, \\ \Pr(\|\mathbf{z}\|_\infty < B) &= \prod_{i=0}^{(k+l)n-1} (1 - \Pr(|z_i| \geq B)) \approx 1 - n(k+l)\Pr(|z_i| \geq B). \end{aligned}$$

We can easily obtain the following corollary.

Corollary 1. Using the same assumptions as in Theorem 3.1, the resulting signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is ε_c -correct, where $\varepsilon_c = 1 - \Pr(\|\mathbf{z}\|_\infty < B)$.

Then we need to prove that the *Rhyme_ID* scheme achieves EU-CMA security. We show that *Rhyme_ID* is a lossy identification scheme and lossy-special

sound. In corollary 2 and 3, we obtain that the signature scheme $\text{FS}[\text{Rhyme_ID}, \text{H}]$ is EU-NMA-secure in the ROM, and in the QROM under some parameters constraint.

Theorem 3.2. *Let $k, l > 0$ and let $B > 0$ and $q > 2B$ be an odd modulus. Then our Rhyme_ID identification protocol:*

- *satisfies key-indistinguishability under $\text{LWE}_{k, l-1, \sigma', q}$ assumption,*
- *is ε_{ls} lossy sound for*

$$\varepsilon_{ls} = \frac{1}{|\mathcal{C}|} + \frac{|\mathcal{C}|^2(2B+1)^{2(k+l)}}{q^k}.$$

- *is also lossy-special-sound, under the $\text{SIS}_{k, k+l, q, 2B}$ assumption.*

Proof. Since the key generation, signing and verification processes in the Rhyme_ID scheme are quite similar to those in the G+G scheme, the lossy key generation, along with the related proofs and analyses, also share similarities.

key-indistinguishability. The lossy key generation algorithm samples $\mathbf{b} \leftarrow U(\mathcal{R}_q^k)$ rather than computes $\mathbf{b} \leftarrow \mathbf{A}_{\text{gen}} \cdot \mathbf{s}_{\text{gen}} + \mathbf{e}_{\text{gen}}$ in the **IGen**. Lossy verification keys are computationally indistinguishable from non-lossy ones, under $\text{LWE}_{k, l-1, \sigma', q}$ assumption.

ε_{ls} -lossy soundness. If the lossy verification key $\mathbf{A}$ satisfies the following condition: For all commitment $\mathbf{w}$, there exists at most one challenge value c such that a response $\mathbf{z}$ exists making $(\mathbf{w}, c, \mathbf{z})$ pass verification, then as the challenge is sampled uniformly and independently of $\mathbf{w}$. The probability that the prover randomly guesses a valid challenge c is at most $\frac{1}{|\mathcal{C}|}$, where $\mathcal{C}$ is the challenge space.

We then focus on proving that the above holds with overwhelming probability over the choice of the lossy key $\mathbf{A}$. If there exists $\mathbf{w}, c_0, c_1, \mathbf{z}_0, \mathbf{z}_1$ with $\|\mathbf{z}_0\|, \|\mathbf{z}_1\| < B$ and $c_0 \neq c_1 \in \mathcal{C}$, such that we have both $\mathbf{A}\mathbf{z}_0 = \mathbf{w} + qc_0\mathbf{j} \pmod{2q}$ and $\mathbf{A}\mathbf{z}_1 = \mathbf{w} + qc_1\mathbf{j} \pmod{2q}$. Then, we have

$$\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = q(c_0 - c_1)\mathbf{j} \pmod{2q}.$$

Recall that $\mathbf{A}$ is of the form $(-2\mathbf{b} + q\mathbf{j}|2\mathbf{A}_{\text{gen}}|2\mathbf{Id}_k)$, with $\mathbf{A}_{\text{gen}}, \mathbf{b}$ uniformly sampled from $\mathcal{R}_q$. Hence, the matrix $\mathbf{A} \pmod{q}$ is of the form $(-\mathbf{b}|\mathbf{A}_{\text{gen}}|\mathbf{Id}_k)$, since q is odd. Then the above implies that $(-\mathbf{b}|\mathbf{A}_{\text{gen}}|\mathbf{Id}_k)(\mathbf{z}_0 - \mathbf{z}_1) = 0 \pmod{q}$ with $\mathbf{z}_0 - \mathbf{z}_1 \neq 0 \pmod{q}$. This happens with probability at most $1/q^k$. To conclude, note that there are at most $(4B+1)^{(k+l)} \cdot |\mathcal{C}|^2$ choices for $\mathbf{z}_0 - \mathbf{z}_1, c_0$ and c_1 . A union bound therefore implies that the probability over $\mathbf{A}$ that there is a commitment with at least two challenges permitting valid transcripts is at most $|\mathcal{C}|^2(4B+1)^{k+l}/q^k$. Our lossy identification scheme is then ε_{ls} -lossy-sound, with

$$\varepsilon_{ls} \leq \frac{1}{|\mathcal{C}|} + \frac{|\mathcal{C}|^2(4B+1)^{k+l}}{q^k},$$

which shows that it is lossy combined with key-indistinguishability.

Lossy special soundness. Assume there exists a PPT adversary $\mathcal{A}$ which, given a lossy verification key $vk = \mathbf{A}$, produces two valid transcripts $(\mathbf{w}, c_0, \mathbf{z}_0)$ and $(\mathbf{w}, c_1, \mathbf{z}_1)$ with $c_0 \neq c_1$. It can be turned into an $\text{SIS}_{k,k+l,q,2B}$ solver. Indeed, by definition, such transcripts satisfy $\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = q(c_0 - c_1)\mathbf{j} \bmod 2q$. Notice that we have $\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = 0 \bmod q$, which implies that $\mathbf{z}_0 - \mathbf{z}_1$ is a solution to the (uniformly sampled) SIS instance defined by $\mathbf{A}$. In addition, when reducing modulo 2, we also have $\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = (c_0 - c_1)\mathbf{j} \bmod 2$, which implies that $\mathbf{z}_0 \neq \mathbf{z}_1$. Finally, note that the condition on B implies that $\|\mathbf{z}_0 - \mathbf{z}_1\| \leq 2B$ (as transcript validity implies $\|\mathbf{z}\| \leq B$), and that $\mathbf{z}_0 - \mathbf{z}_1 \neq 0 \bmod q$. Hence, there exists an adversary $\mathcal{B}$ against the $\text{SIS}_{k,k+l,q,2B}$ problem such that:

$$\text{Adv}(\mathcal{A}) \leq \text{Adv}^{\text{SIS}_{k,k+l,q,2B}}(\mathcal{B}),$$

which completes the proof of the theorem.

Building on this foundation, we can derive the corollaries during the reduction process.

Corollary 2. Using the same assumptions as in Theorem 3.2, the resulting signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is EU-NMA-secure in the ROM. Namely, for any adversary $\mathcal{A}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, H]$, there exists an adversary $\mathcal{B}$ against the $\text{SIS}_{k,k+l,q,2B}$ assumption as well as an adversary $\mathcal{B}'$ against the $\text{LWE}_{k,l-1,\sigma',q}$ assumption such that

$$\text{Adv}^{\text{EU-NMA}}(\mathcal{A}) \leq Q_H \cdot \left(\sqrt{\text{Adv}^{\text{SIS}_{k,k+l,q,2B}}(\mathcal{B})} + \frac{2}{|\mathcal{C}|} \right) + \text{Adv}^{\text{LWE}_{k,l-1,\sigma',q}}(\mathcal{B}').$$

Corollary 3. Using the same assumptions as in Theorem 3.2, and if ε_{ls} is negligible, the signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is EU-NMA-secure, in the QROM. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, H]$ making at most Q_H (possibly quantum) hash queries, there exists a quantum adversary $\mathcal{B}$ against the $\text{LWE}_{k,l-1,\sigma',q}$ assumption such that:

$$\text{Adv}^{\text{EU-NMA}}(\mathcal{A}) \leq \text{Adv}^{\text{LWE}_{k,l-1,\sigma',q}}(\mathcal{B}) + 8(Q_H + 1)^2 \cdot \left(\frac{1}{|\mathcal{C}|} + \frac{|\mathcal{C}|^2(2B+1)^{2(k+l)}}{q^k} \right)$$

To conclude this section, we introduce an additional assumption of a similar flavor as the SelfTargetMSIS assumption [19], which allows to directly prove EU-NMA-security of $\text{FS}[\text{Rhyme_ID}, H]$ in the QROM as it is (up to LWE) the EU-NMA security game of the resulting signature. As for SelfTargetMSIS, this problem can be related in the ROM to SIS, using the special soundness property of the scheme.

Definition 3.1. (*SelfTargetSIS*) Let $k, l > 0$, $B > 0$ and $q > 2B$ be an odd modulus. The TargetSIS _{m,k,ℓ,B,q} states that given a matrix $\mathbf{A} := (-2\mathbf{b} + q\mathbf{j} \mid 2\mathbf{A}_{\text{gen}} \mid 2\mathbf{I}_k)$,

where $\mathbf{A}_{gen} \leftarrow U(\mathcal{R}_q^{k \times (l-1)})$, and oracle access to a hash function H , it is computationally hard to find $c \in \mathcal{C}$, $\mathbf{z} \in \mathbb{Z}_{2q}^k$ and $\mu \in \{0, 1\}^*$ such that $H(\mathbf{A}\mathbf{z} - qc\mathbf{j}, \mu) = c$ and $\|\mathbf{z}\| < B$.

By leveraging the previous theorems and conclusions, we have already established that our scheme achieves EU-NMA security. Next, we need to prove that the Rhyme_ID protocol is HVZK and has a large commitment min-entropy. From this, we can further deduce that our signature scheme FS[Rhyme_ID, H] is EU-CMA-secure.

Theorem 3.3. *Let $k, l > 0$ and let $\gamma > 0$ and q be an odd modulus. Then the Rhyme_ID identification protocol satisfies HVZK and its commitment min-entropy is $\geq k \log(8/3)$.*

Proof. We prove both properties separately.

HVZK. The simulator on input a challenge $c \in \mathcal{C}$ and a public matrix $\mathbf{A}$ samples $\mathbf{z}$ according to the distribution $\chi(0, 1) + \chi(0, 1) + \sum^{\tau} \chi(0, 2)$ (except for z_0 who has an extra $\xi(c)$), sets $\mathbf{w} = \mathbf{A}\mathbf{z} - qc\mathbf{j}$ and returns $(\mathbf{w}, c, \mathbf{z})$ as a transcript. Based on the previous description of the sampling process, it is easy to observe that the distribution generated by the simulator is exactly the same with the target distribution. Consequently, the statistical distance and Rényi divergence between the simulated distribution and the target distribution can be computed and shown to be 0 and 1 respectively, thereby ensuring the HVZK property.

Commitment Min-Entropy. Let $\mathbf{w} \in \mathbb{Z}_{2q}^{k+l}$ and $(Y_1^\top, Y_2^\top)^\top \leftarrow D(\mathbf{Y})_{\mathbb{Z}_{2q}^{k+l}}$, where Y_1 takes values in $\mathbb{Z}_{2q}^l$. Given a matrix $\mathbf{A} = (\mathbf{A}_0 \mid 2\mathbf{Id}_k) \in \mathbb{Z}_{2q}^{k \times (k+l)}$, it holds that

$$\begin{aligned} \Pr_{(Y_1, Y_2)} [\mathbf{A}_0 Y_1 + 2Y_2 = \mathbf{w} \bmod 2q] &= \Pr_{(Y_1, Y_2)} [2Y_2 = \mathbf{w} - \mathbf{A}_0 Y_1 \bmod 2q] \\ &\leq \Pr_{(Y_1, Y_2)} [Y_2 = (\mathbf{w} - \mathbf{A}_0 Y_1) \zeta \bmod q] \end{aligned}$$

where ζ is the modular inverse of $2 \bmod q$. Hence, the min-entropy of the commitment is $\geq H_\infty(Y_2 \bmod q \mid Y_1)$. Let $\mathbf{t} \in \mathbb{Z}^k$ be the value that maximizes the probability for Y_2 . Since the y_i in our sampling scheme are independent from each other and follows $\chi(0, 1) + \chi(0, 1) + \sum^{\tau/2} \chi(0, 2)$, we will omit the last part for simplicity and show that the distribution of $\chi(0, 1) + \chi(0, 1)$ can provide enough entropy. As each slot of Y_2 is sampled from $\chi(0, 1) + \chi(0, 1)$, so the support of Y_2 is $\{-2, -1, 0, 1, 2\}^k$ and we have:

$$\Pr_{Y_2|Y_1=y_1}(Y_2 = \mathbf{t}) = \Pr(Y_2 = 0^k) \leq (8/3)^{-k}.$$

By further analyzing the relationship between $(\mathbf{w}, \mathbf{z}, c)$ and y , we can ultimately conclude that:

$$H_\infty(w|c) \geq H_\infty(Y_2 \bmod q \mid Y_1) = - \sum_{Y_2} \Pr(Y_2 = \mathbf{t}) \cdot \log(\Pr(Y_2)) \geq k \log(8/3) \approx 1.4k$$

We then obtain the following corollary.

Corollary 4. Using the same assumptions as in Theorem 3.3 the resulting signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is EU-CMA-secure in the QROM, provided it is EU-NMA-secure. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the EU-CMA security of $\text{FS}[\text{Rhyme_ID}, H]$ making at most Q_S (classical) sign queries and at most Q_H (possibly quantum) hash queries, there exists an adversary $\mathcal{B}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, H]$ such that:

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) &\leq (1 - 0)^{Q_S} \text{Adv}^{\text{EU-NMA}}(\mathcal{B}) \\ &\quad + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot (8/3)^{-k}} \end{aligned}$$

Finally, in order to show the strong unforgeability of the scheme, we show that Rhyme_ID has Computational Unique Response (CUR).

Theorem 3.4. Let $k, l > 0$ and let $B > 0$ and $q > 2B$ be an odd modulus. Then the Rhyme_ID identification protocol satisfies the CUR property under both SIS and LWE assumptions. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the CUR property, there exists an adversary $\mathcal{B}$ against $\text{LWE}_{k, l-1, \sigma', q}$ and an adversary $\mathcal{B}'$ against $\text{SIS}_{k, k+l, q, 2B}$ such that:

$$\text{Adv}^{\text{CUR}}(\mathcal{A}) \leq \text{Adv}^{\text{LWE}_{k, l-1, \sigma', q}}(\mathcal{B}) + \text{Adv}^{\text{SIS}_{k, k+l, q, 2B}}(\mathcal{B}').$$

Proof. We first change the verification key to the lossy one, as defined in the proof of Theorem 3.3. This means that the matrix $2^{-1}\mathbf{A}$ is now uniform modulo q among matrices in Hermite Normal Form. Now, given elements $(\mathbf{w}, c, \mathbf{z}_0, \mathbf{z}_1)$ such that $\mathbf{A}\mathbf{z}_0 - qc\mathbf{j} = \mathbf{A}\mathbf{z}_1 - qc\mathbf{j} \pmod{2q}$, we see that $\mathbf{A}(\mathbf{z}_0 - \mathbf{z}_1) = \mathbf{0} \pmod{q}$. This means that breaking the CUR property of the scheme implies finding a SIS solution for the instance defined by $2^{-1}\mathbf{A} \pmod{q}$.

Corollary 5. Using the same assumptions as in Theorem 3.4 the resulting signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is sEU-CMA-secure in the QROM, provided it is EU-NMA-secure. Namely, for any (possibly quantum) adversary $\mathcal{A}$ against the EU-CMA security of $\text{FS}[\text{Rhyme_ID}, H]$ making at most Q_S (classical) sign queries and at most Q_H (possibly quantum) hash queries, there exist an adversary $\mathcal{B}$ against the EU-NMA security of $\text{FS}[\text{Rhyme_ID}, H]$, an adversary $\mathcal{B}'$ against $\text{LWE}_{k, l, \sigma', q}$ and an adversary $\mathcal{B}''$ against $\text{SIS}_{k, k+l, q, 2B}$ such that:

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) &\leq 1^{Q_S} \text{Adv}^{\text{EU-NMA}}(\mathcal{B}) \\ &\quad + 3Q_S/2 \cdot \sqrt{(Q_H + Q_S + 1) \cdot (8/3)^{-k}} + Q_S \cdot (8/3)^{-k} \\ &\quad + \text{Adv}^{\text{LWE}_{k, l, \sigma', q}}(\mathcal{B}') + \text{Adv}^{\text{SIS}_{k, k+l, q, 2B}}(\mathcal{B}'') \end{aligned}$$

Building upon the above corollaries, which are derived from the properties of

our identification protocol, we can get the following theorem that the signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is EU-CMA in the ROM and in the QROM, by applying the Fiat-Shamir transform.

Theorem 3.5. *Let $k, l > 0$ and let $B > 0$ and $q > 2B$ be an odd modulus. Then the signature scheme $\text{FS}[\text{Rhyme_ID}, H]$ is:*

- *EU-CMA-secure in the ROM under the assumption. Namely, for any adversary $\mathcal{A}$ against the EU-CMA security of making at most Q_s sign queries and at most Q_h hash queries, there exists an adversary $\mathcal{B}$ against the $\text{SIS}_{k,k+l,q,2B}$ assumption and an adversary $\mathcal{B}'$ against the $\text{LWE}_{k,l-1,\sigma',q}$ assumption such that:*

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) \leq & 1^{Q_s} \left[Q_h \cdot \left(\sqrt{\text{Adv}^{\text{SIS}_{k,k+l,q,2B}}(\mathcal{B})} + \frac{2}{|\mathcal{C}|} \right) \right] \\ & + 3Q_s/2 \cdot \sqrt{(Q_h + Q_s + 1) \cdot (8/3)^{-k}} \\ & + \text{Adv}^{\text{LWE}_{k,l-1,\sigma',q}}(\mathcal{B}'); \end{aligned}$$

- *EU-CMA-secure in the QROM under the $\text{LWE}_{k,l-1,\sigma',q}$ assumption, assuming is negligible. Namely, for any quantum adversary against the EU-CMA security of making at most classical sign queries and at most quantum hash queries, there exists an adversary against the assumption such that:*

$$\begin{aligned} \text{Adv}^{\text{EU-CMA}}(\mathcal{A}) \leq & 1^{Q_s} \text{Adv}^{\text{LWE}_{k,l-1,\sigma',q}}(\mathcal{B}') \\ & + 1^{Q_s} \cdot 8(Q_h + 1)^2 \cdot \left(\frac{1}{|\mathcal{C}|} + \frac{|\mathcal{C}|^2(2B+1)^{2(k+l)}}{q^k} \right) \\ & + 3Q_s/2 \cdot \sqrt{(Q_h + Q_s + 1) \cdot (8/3)^{-k}}. \end{aligned}$$

Moreover, these two bounds holds when $\mathcal{A}$ is an adversary against the sEU-CMA security of the scheme by adding an extra $+Q_s \cdot (8/3)^{-k} + \text{Adv}^{\text{LWE}_{k,l-1,\sigma',q}}(\mathcal{B}') + \text{Adv}^{\text{SIS}_{k,k+l,q,2B}}(\mathcal{B})$ term on the right hand side.

3.7 Cost of known attacks

For the concrete security analysis, we list the best known lattice attacks and consider their costs for attacking our scheme. All the best known attacks rely on the Block–Korkine–Zolotarev (BKZ) lattice reduction algorithm [11, 27, 38]. The BKZ algorithm is a lattice basis reduction algorithm that repeatedly uses a Shortest Vector Problem (SVP) solver in small-dimensional projected sublattices. The dimension b of these projected sublattices is called the block-size. BKZ with block-size b hence relies on an SVP solver in dimension b . The block-size drives the cost of BKZ and determines the resulting basis’s quality. It provides a quality/time trade-off: If b gets larger, better quality will be guaranteed, but

the time complexity for the SVP solver will exponentially increase. The time complexity of the b -BKZ algorithm is the same as the SVP solver for dimension b , up to polynomial factors. Hence the time complexity differs depending on the SVP solver used. The most efficient SVP algorithm uses the sieving method proposed by Becker et al. in [5] which takes time $\approx 2^{0.292b+o(b)}$. The fastest known quantum variant is proposed by Chailloux and Loyer in [10] and takes time $\approx 2^{0.257b+o(b)}$.

Based on the BKZ algorithm, we will follow the core-SVP methodology from [36] and as in the subsequent lattice-based schemes [1, 22, 24, 31, 35]. It is regarded as a conservative way to set security parameters. We ignore the polynomial factors and the $o(b)$ terms in the exponents of the run-time bounds above for the time complexity of the BKZ algorithm. We consider the primal attack and the dual attack for MLWE, and the plain BKZ attack for MSIS and Bi-modalSelfTargetMSIS problems. We remark that any $MLWE_{n,q,k,l,\sigma_1}$ instance can be viewed as an $LWE_{q,nk,nl,\sigma'}$ instance, and also any $MSIS_{n,q,k,l,B}$ can be viewed as an $SIS_{q,nk,nl,B}$ instance. Even though the MLWE and MSIS problems have some extra algebraic structure compared to the LWE and SIS problems, we do not currently know how to exploit it to improve the best known attacks. For this reason, we estimate the concrete hardness of the MLWE and MSIS problems over the structured lattices as the concrete hardness of the corresponding binary-LWE and SIS problems over the unstructured lattices.

We summarize the costs of the known attacks in Table 4. In the table, the required block-sizes for BKZ and the costs of the attacks in core-SVP hardness are given, estimated by the better result from the estimator proposed in G+G and the Albrecht's lattice-estimator. The parameters for MLWE and MSIS problems are chosen based on the above analysis. The numbers in parentheses are for the SUF-CMA security of SIS problem.

Primal attack. Given an LWE instance $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$, we first define the lattices $\Lambda_m = \{\mathbf{v} \in \mathbb{Z}^{l+m+1} : \mathbf{B}\mathbf{v} = \mathbf{0} \bmod q\}$ for all $m \leq k$, where $\mathbf{B} = (\mathbf{A}_{[m]} | \mathbf{I}_m | \mathbf{b}_{[m]}) \in \mathbb{Z}^{m \times (l+m+1)}_q$, $\mathbf{A}_{[m]}$ is the uppermost $m \times l$ sub-matrix of $\mathbf{A}$ and $\mathbf{b}_{[m]}$ is the uppermost m -dimensional sub-vector of $\mathbf{b}$. As $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$ is an LWE instance, there exist $\mathbf{s}$ and $\mathbf{e}$ short such that $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$. This implies that $(\mathbf{s} | \mathbf{e} | -1)$ is a short vector of Λ_m . The primal attack consists in running BKZ on Λ_m to find short vectors. The variable m is optimized to minimize the cost of the attack.

Dual attack. Given an LWE instance $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{k \times l} \times \mathbb{Z}_q^k$, we first define the lattices $\Lambda'_m = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^m \times \mathbb{Z}^l : \mathbf{A}_{[m]}^T \mathbf{u} + \mathbf{v} = \mathbf{0} \bmod q\}$ for all $m \leq k$, where $\mathbf{A}_{[m]}$ is the uppermost $m \times l$ sub-matrix of $\mathbf{A}$. If $(\mathbf{u}, \mathbf{v})$ is a short vector in Λ'_m , then $\mathbf{u}^T \mathbf{b} = \mathbf{v}^T \mathbf{s} + \mathbf{u}^T \mathbf{e}_{[m]}$ is short if $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ for short vectors $\mathbf{s}$ and $\mathbf{e}$, and is uniformly distributed modulo q if $\mathbf{b}$ is uniform and independent from $\mathbf{A}$ (here $\mathbf{e}_{[m]}$ refers to the uppermost m -dimensional sub-vector of $\mathbf{e}$). This provides a

distinguishing attack. The dual attack consists in finding a short non-zero vector in the lattice Λ'_m using BKZ. The variable m is optimized to minimize the cost of the attack.

Hybrid Dual Attack. Our scheme is based on binary-LWE, for which the hybrid attack approach is typically effective. A hybrid attack guesses part of the secret key and then seeks a trade-off between the cost of guessing and the cost of the dual attack. Among the guessing techniques, meet-in-the-middle (MITM) proposed by [33] works with the best complexity. And for hybrid dual attack, [6] gets the optimal result. However, there is still doubt about whether exponentially number of short vectors can be obtained by sieving algorithm under the condition that some of the latest sieving improvements are incompatible with it according to [35]. As a result, we list the corresponding result of [6] separately as a reference, under core-SVP with the assumption proposed in [17].

Table 3. Hybrid Dual Attack for $q = 257$

Security of HYBRID 2M [6]	128	192	256
BKZ block-size b	502	785	1070
Classical hardness	142	221	301
Quantum hardness	129	202	275

SIS attack. To analyze the hardness of BimodalSelfTargetMSIS, we analyze the hardness of the corresponding MSIS. Intuitively, if we assume that H is a cryptographic hash, then the input structure will not help find the preimage. So we can assume that M is fixed. Then the problem turns into finding the preimage $\mathbf{x}$ of c with respect to $H(\cdot, M)$ and then finding $\mathbf{z}'$ satisfying $\mathbf{x} = \mathbf{A}\mathbf{z}' - qc\mathbf{j} \bmod 2q$. Apart from the first step, if we have the preimage c , then the second step will be turned into finding $\mathbf{y}'$ satisfying $(-\mathbf{2b}|\mathbf{A}_0|\mathbf{Id}_k) \cdot \mathbf{y}' = \mathbf{t} \bmod \mathbf{q}$, for a known vector $\mathbf{t}$ over $\mathcal{R}_q$. Here, $\mathbf{y}'$ is defined as $\mathbf{y}' = ((\mathbf{y}_0 - \mathbf{x}'_0)/2, \mathbf{y}_1, \dots, \mathbf{y}_{k+l-1})^T$ and $t = 2^{-1} \cdot \mathbf{x} + \mathbf{x}'_0 \mathbf{b} \bmod q$, where $\mathbf{x}'_0 = (\mathbf{x}_0 + \mathbf{c} \bmod 2)$ which actually decides the LSB of $\mathbf{y}_0$. Also, $\|\mathbf{y}'\|_2$ is bounded by the same bound used for $\|\mathbf{y}\|_2$. This implies that solving BimodalSelfTargetMSIS is at least as hard as solving MSIS with the same norm bound or finding the preimage of a hash as an attack perspective. However, for the hardness of BimodalSelfTargetMSIS problem, we will analyze it more conservatively, as the hardness of MSIS problem with a twice larger norm bound, taking into account the classical reduction from MSIS to BimodalSelfTargetMSIS in Theorem 2.1. We analyze the best known attacks for SIS problem, for both MSIS and BimodalSelfTargetMSIS problems that the unforgeability of our signature scheme relies on.

Given an SIS instance $\mathbf{A} \in \mathbb{Z}_q^{k \times l}$ with a bound B , we define the lattices $\Lambda''_m = \{\mathbf{u} \in \mathbb{Z}^m : \mathbf{B}\mathbf{u} = \mathbf{0} \bmod q\}$ for all $m \leq k + l$, where $\mathbf{B}$ is the $k \times m$

leftmost sub-matrix of $(\mathbf{A}|\mathbf{I}_k)$. Then a short non-zero vector in the lattice Λ''_m is a solution to the SIS problem. Once more, we use BKZ and optimize the choice of m . Note that if $B > q$, then there are some trivial non-zero solutions to SIS problem such as $(q, 0, \dots, 0)$ with $l2 - norm < B$. Depending on the parameters, the security could be affected by some existing attacks [31]. We choose the prime q larger than the MSIS bound B to avoid such weaknesses.

Ratio attack. Recently, a ratio attack has been proposed to attack G+G in [40], whose main idea is to use the truth that $s_0 = 1 \in R_q$ is different from other parts of $\mathbf{S}$. So the distribution of z_0 are also different from other parts of $\mathbf{z}$. Besides, $z_{0,0}, z_{i,j}$ share the same masking element $\gamma_0 = x_0 + c_0$, this makes $\frac{z_{i,j}}{z_{0,0}}$ a Cauchy distribution who is related to $s_{i,j}$ and γ_0 since other γ_i and y_i are all independent of each other. More specifically, the distribution of $\frac{z_{i,j}}{z_{0,0}}$ is a Cauchy distribution $C(\alpha_{i,j}, \beta_{i,j})$, where $\alpha_{i,j} = \rho_{i,j} \frac{\sigma_{z_{i,j}}}{\sigma_{z_{0,0}}}$, $\beta_{i,j} = \frac{\sigma_{z_{i,j}}}{\sigma_{z_{0,0}}} \sqrt{1 - \rho_{i,j}^2}$ and $\rho_{i,j} = \frac{\mathbb{E}((Z_{i,j} - \mu_{Z_{i,j}})(Z_{0,0} - \mu_{Z_{0,0}}))}{\sigma_{Z_{i,j}} \sigma_{Z_{0,0}}}$. Since the $z_{0,0}, z_{i,j}$ share the same masking element $\gamma_0 = x_0 + c_0$, $\rho_{i,j} = \frac{s_{i,j} \gamma_0^2}{\sigma_{Z_{i,j}} \sigma_{Z_{0,0}}}$.

To avoid ratio attack, we introduce $\gamma' = x' + c'$ to share through each element of $\mathbf{z}$ as part of $\mathbf{y}_2$. As a result, in this case $\rho_{i,j} = \frac{s_{i,j} \gamma_0^2 + \sum_n s'_{i,k_1} s'_{0,k_2} \gamma_{k_3}^2}{\sigma_{Z_{i,j}} \sigma_{Z_{0,0}}}$, where $\mathbf{s}'$ and γ' are all random sampled, thus covering the information of the secret key against ratio attack.

3.8 Parameter Sets

We propose three different parameter sets with varying security levels in Table 4. The ring dimension n and the modulus q are set to 256 and 257 across all parameter sets. Our choice of modulus q allows for high-efficient NTT computations.

The variable B denotes the verification bound. It should be noted that in our scheme, we don't rely on verification bound to achieve HVKZ like rejection sampling. However, setting a verification bound provides a trade-off between sign efficiency and signature size. Besides, a verification bound is also convenient for estimating security of SIS problem. As a result, we set it significantly smaller than q to avoid potential attacks exploiting the q -vectors. Moreover, we also try to use it to achieve a compact signature size with an acceptable repetition rate.

The numbers in parentheses for SIS security are for the strongly-unforgeable (i.e. hard to come up with a second distinct signature for an already-seen message/signature pair) version of the signature scheme. The C_Rate in the computation of signature size denotes the compress rate of rANS, which is about 0.85-0.9 according to our experiments.

Table 4. Parameters for $q = 257$

Security	128	192	256
q	257	257	257
(k, l)	(2, 3)	(3, 4)	(4, 5)
B	32	48	60
τ	30	60	128
challenge entropy $[\log_{\tau}^{(256)}]$	129	197	251
$Pr(\mathbf{z} _{\infty} < B)$	0.91	0.95	0.58
Forgery			
BKZ block-size b	483(371)	668(527)	862(691)
Classical hardness	141(108)	195(154)	252(202)
Quantum hardness	124(95)	171(135)	221(177)
Key-Recovery			
BKZ block-size b	507	712	983
Classical hardness	148	228.9	305.0
Quantum hardness	130.3	201.5	268.4
Signature size: $(k + l) * 256 * \lceil \log_2(2B - 1) \rceil / 8 * C_Rate + 256 * \log(2) / 8$	834	1312	1867
Public key size: $\lceil k * 256 * 8.008 / 8 \rceil + 256 * \log(2) / 8$	545	801	1058
Secret key size: $(2k + 2l - 1 + 1) * 256 * \log_2(2) / 8 + 256 * \log(2) / 8$	352	480	608

3.9 Performance Analysis

In this subsection, we report the performance of the implementation. In Tables 5 and 6, we present the performance and size results of HAETAE, Dilithium and Rhyme respectively. All benchmarks were obtained on one core of an Intel Core i9-12900, with TurboBoost and hyperthreading disabled. All cycle counts reported are the median and average of the cycle counts of 1,000 executions of the respective functions. It is seen that our scheme achieves a compact signature with high efficiency.

4 Conclusion

In this paper, we propose a new post-quantum lattice-based digital signature which is based on Fiat-Shamir structure and 3C sampling method. We prove its security in ROM/QROM, and provide concrete parameters as well as reference implementation to show that the scheme enjoys high efficiency and very small signature size compared with former results.

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Parameter set		KeyGen	Sign	Verify
Rhyme-128	med	104513	315642	140183
	avg	104496	314976	140775
Rhyme-192	med	185666	491636	241427
	avg	185692	495818	243262
Rhyme-256	med	286506	1245400	372015
	avg	293134	1251825	375161
HAETAE-120	med	496333	1688959	114705
	avg	654722	2388867	115286
HAETAE-180	med	698998	2507237	210296
	avg	1009883	3297772	211484
HAETAE-260	med	691088	3181843	264210
	avg	895996	4536198	264401
Dilithium-120	med	118348	384538	123745
	avg	118645	497856	125074
Dilithium-180	med	217724	651107	196512
	avg	218284	812481	197122
Dilithium-260	med	320536	851463	324732
	avg	321979	1022420	326667

Table 5. Median and average cycle counts of 1000 executions for Rhyme, HAETAE and Dilithium

Parameter set	Signature	Public key	Secret key	SIG+PK
Rhyme-128	834	545	897	1379
Rhyme-192	1312	801	1281	2113
Rhyme-256	1867	1058	1666	2925
HAETAE-120	1,474	992	1,408	2466
HAETAE-180	2,349	1,472	2,112	3821
HAETAE-260	2,948	2,080	2,752	5028
Dilithium-120	2,420	1,312	2,528	3732
Dilithium-180	3,293	1,952	4,000	5245
Dilithium-260	4,595	2,592	4,864	7187

Table 6. Signature and key sizes (bytes) of Rhyme, HAETAE and Dilithium.

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